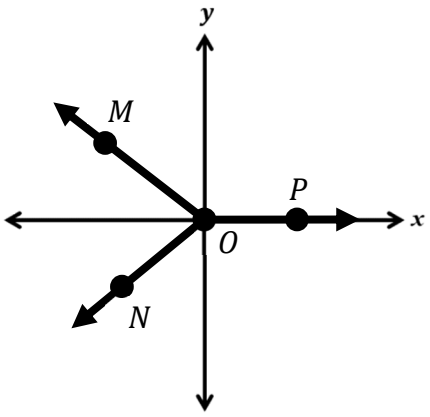
3.2A Radians

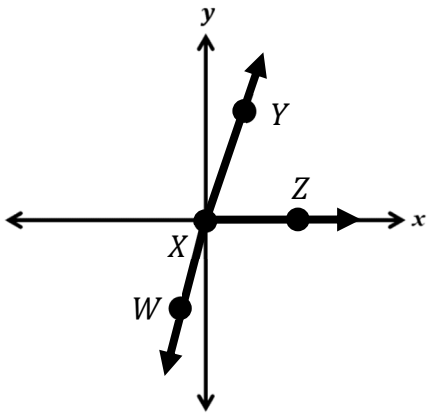
AP Precalculus Name: _____ **CA #1**

For each set of axes, name all the angles that are in standard position. Give the initial ray and terminal ray of each angle.

1.



2.



The measurement of an angle in standard position is listed. In which quadrant is the terminal ray?

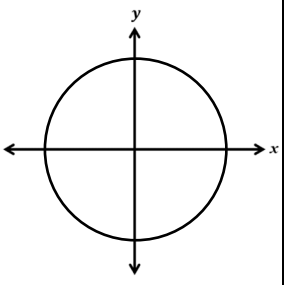
3. 4.6π

4. 7.3π

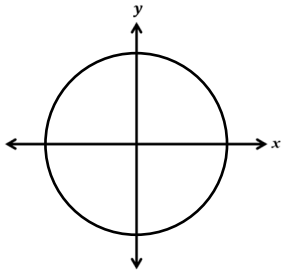
5. -5.6π

Below are various measurements of a circle's radius, an angle within the circle, or the arc subtended by the angle. SKETCH the approximate angle on the axes and find the missing value.

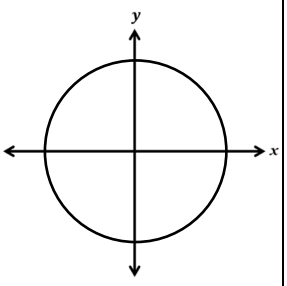
6. Radius is 9.71 and the length of an arc subtended by an angle is 16.9. Find the measure of the angle.



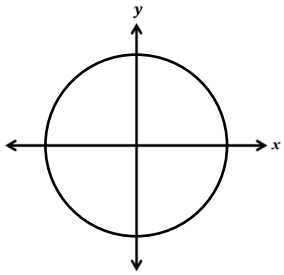
7. Radius is 9.63 and an angle is 0.98π radians. Find the length of the arc subtended by the angle.



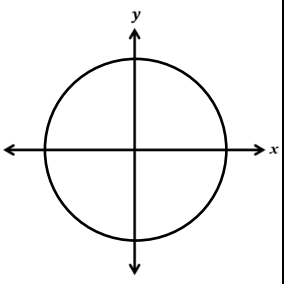
8. An angle is 1.4π radians and the length of an arc subtended by the angle is 15.8. What is the radius of the circle?



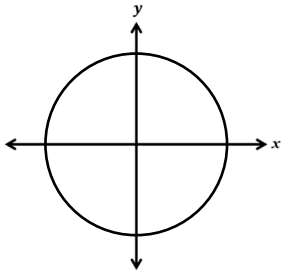
9. Radius is 4.31 and an angle is 4.47 radians. Find the length of the arc subtended by the angle.



10. An angle is 0.78 radians and the length of an arc subtended by the angle is 2.09. What is the radius of the circle?



11. Radius is 5.1 and the length of an arc subtended by an angle is 26. Find the measure of the angle.



Answers to 3.2A CA #1

1. Angle POM with initial ray $\overrightarrow{OP}$ and terminal ray $\overrightarrow{OM}$ Angle PON with initial ray $\overrightarrow{OP}$ and terminal ray $\overrightarrow{ON}$		2. Angle ZXY with initial ray $\overrightarrow{XZ}$ and terminal ray $\overrightarrow{XY}$ Angle ZXW with initial ray $\overrightarrow{XZ}$ and terminal ray $\overrightarrow{XW}$		
3. Quadrant II	4. Quadrant III	5. Quadrant I	6. 1.74 radians	7. 29.648
8. 3.592	9. 19.2657	10. 2.679	11. 5.098 radians	

Write your questions and thoughts here!

3.2B Sine, Cosine, and Tangent

3.2B Notes

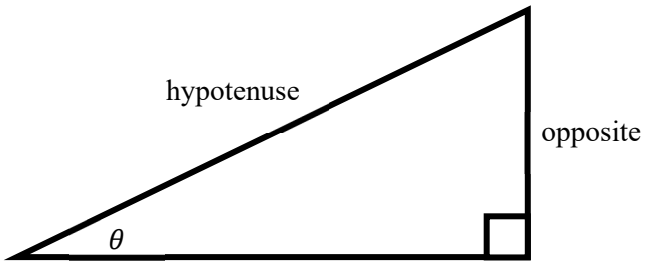
How do you pronounce the following names?

Mr. Jones Mrs. Smith

An abbreviation does not mean we pronounce things differently. It just helps us write it faster. That is the same for sine, cosine, and tangent. How do you pronounce the following?

$\sin\theta$ **$\cos\theta$** **$\tan\theta$**

Recall: SOH – CAH – TOA

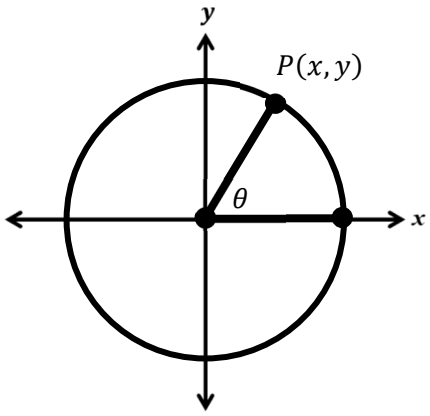


$\sin\theta =$ ———

$\cos\theta =$ ———

$\tan\theta =$ ———

Given an angle in standard position and a circle centered at the origin, there is a point, P, where the terminal ray intersects the circle (see figure to the right).



Using SOH – CAH – TOA, we can find the value of sine, cosine, and tangent of the angle.

$\sin\theta$

The *sine* of the angle is the ratio of the vertical displacement of P from the x-axis to the distance between the origin and point P.

$$\sin\theta = \frac{\text{the vertical displacement of } P \text{ from the } x \text{ axis}}{\text{the distance between the origin and point } P}$$

$$\sin\theta =$$

$\cos\theta$

The *cosine* of the angle is the ratio of the horizontal displacement of P from the y-axis to the distance between the origin and point P.

$$\cos\theta = \frac{\text{the horizontal displacement of } P \text{ from the } y \text{ axis}}{\text{the distance between the origin and point } P}$$

$$\cos\theta =$$

$\tan\theta$

The *tangent* of the angle is the slope, if it exists, of the terminal ray.

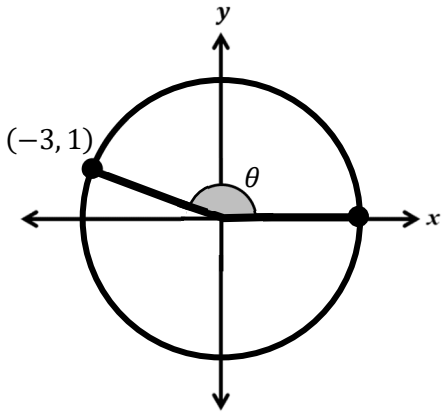
$$\tan\theta = \frac{\text{the vertical displacement of } P \text{ from the } x \text{ axis}}{\text{the horizontal displacement of } P \text{ from the } y \text{ axis}}$$

$$\tan\theta = \quad =$$

Write your questions and thoughts here!

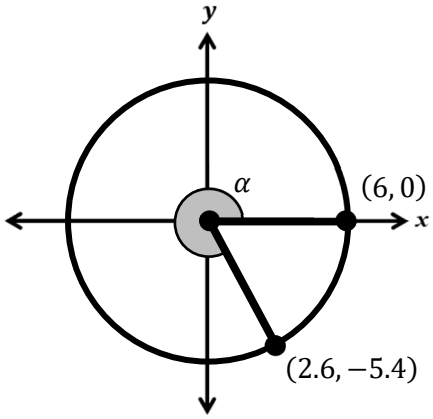
Each figure below gives a circle in the xy-plane with center at the origin, and an angle θ in standard position. Find the value of each expression.

1. Find $\sin\theta$.
2. Find $\cos\theta$.
3. Find $\tan\theta$.



---

4. Find $\sin\alpha$.
5. Find $\cos\alpha$.
6. Find $\tan\alpha$.

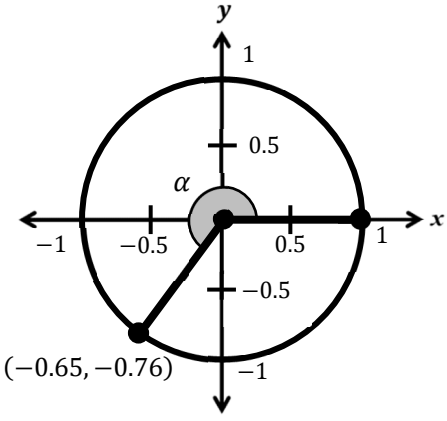


When the circle is a **UNIT CIRCLE** ($r = 1$), the trig values can be simplified.

$\boldsymbol{\sin\theta =}$ $\boldsymbol{\cos\theta =}$ $\boldsymbol{\tan\theta =}$

Each figure below gives a circle in the xy-plane with center at the origin, and an angle θ in standard position. Find the value of each expression.

7. Find $\sin\alpha$.
8. Find $\cos\alpha$.
9. Find $\tan\alpha$.



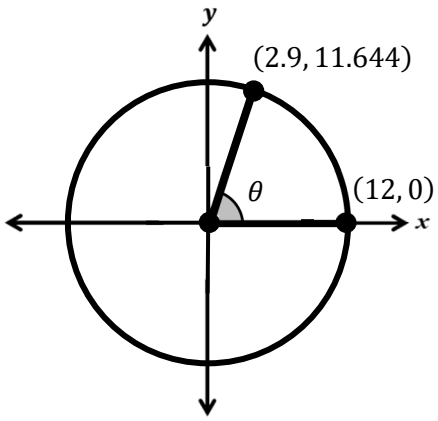
3.2B Sine, Cosine, and Tangent

AP Precalculus

3.2B Practice

Each figure below gives a circle in the xy-plane with center at the origin, and an angle θ in standard position. Find the value of each expression.

1.

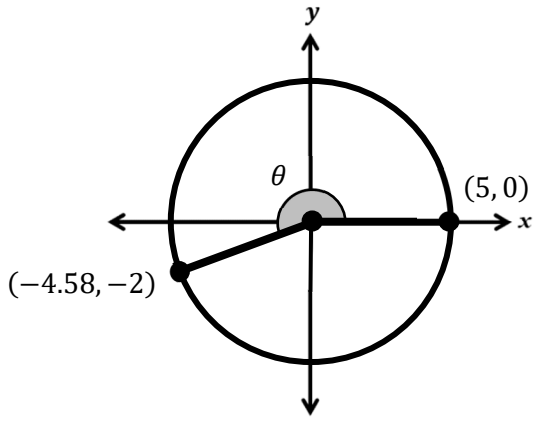


a. $\sin\theta =$

b. $\cos\theta =$

c. $\tan\theta =$

2.

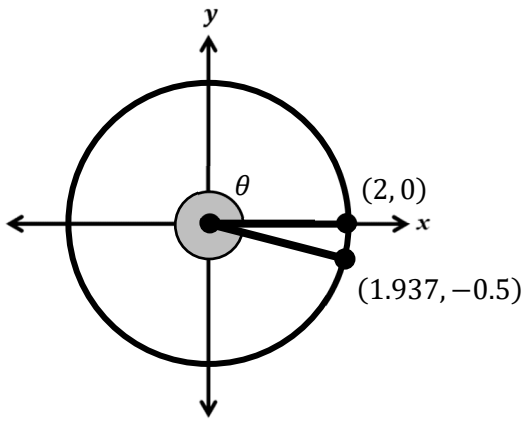


a. $\sin\theta =$

b. $\cos\theta =$

c. $\tan\theta =$

3.

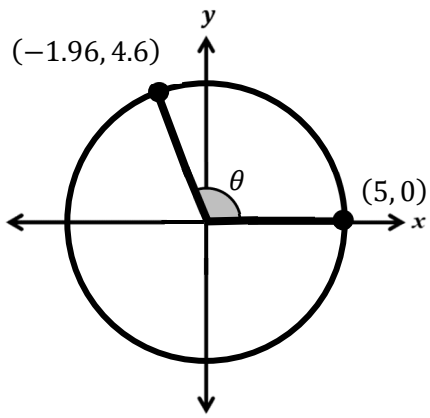


a. $\sin\theta =$

b. $\cos\theta =$

c. $\tan\theta =$

4.

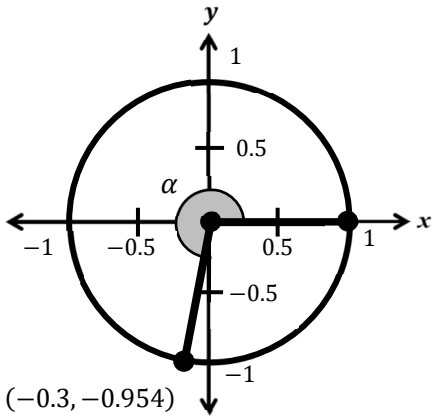


a. $\sin\theta =$

b. $\cos\theta =$

c. $\tan\theta =$

5.

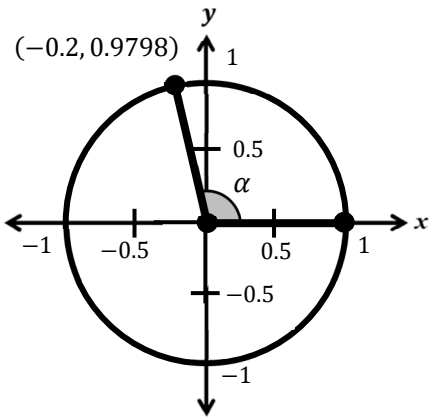


a. $\sin\alpha =$

b. $\cos\alpha =$

c. $\tan\alpha =$

6.



a. $\sin\alpha =$

b. $\cos\alpha =$

c. $\tan\alpha =$

3.2B Sine, Cosine, and Tangent

3.2B Test Prep

7. In the xy-plane, angle ABC is an angle in standard position with terminal ray BC, which intersects the unit circle at the point with coordinates $(0.6, -0.8)$. Which of the following descriptions is correct?

   (A) The sine of angle ABC is $-\frac{4}{3}$.

   (B) The sine of angle ABC is $-\frac{3}{4}$.

   (C) The sine of angle ABC is 0.6.

   (D) The sine of angle ABC is -0.8.

---

8. An angle θ is in standard position in the xy-plane. On the interval $0 \le \theta \le 2\pi$ (one full circle), in which quadrant(s) would the terminal ray of the angle be located for each statement?

   a. $\sin\theta < 0$

   b. $\cos\theta > 0$

   c. $\tan\theta > 0$

9. An angle θ is in standard position in the xy-plane. Which of the following is true about θ on the interval $0 \leq \theta \leq 2\pi$ if $\cos\theta < 0$?

(A) There is no value of θ on $0 \leq \theta \leq 2\pi$ for which $\cos\theta < 0$.

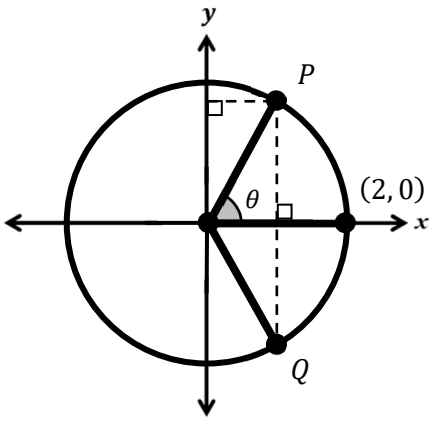
(B) There are values of θ on $0 \leq \theta \leq 2\pi$ for which $\cos\theta < 0$ in all four Quadrants.

(C) There is a value of θ on $0 \leq \theta \leq 2\pi$ for which $\cos\theta < 0$ in Quadrant II only.

(D) There are values of θ on $0 \leq \theta \leq 2\pi$ for which $\cos\theta < 0$ in Quadrants II and III only.

---

10. The figure shows a circle centered at the origin with an angle of measure θ radians in standard position. The terminal ray of the angle intersects the circle at point P, and point Q also lies on the circle. The coordinates of P are (x, y) and the coordinates of Q are $(x, -y)$. Which of the following is true about the cosine of θ?



(A) $\cos\theta = \frac{x}{2}$, because it is the ratio of the horizontal displacement of P from the y-axis to the distance between the origin and P.

(B) $\cos\theta = \frac{-y}{2}$, because it is the ratio of the vertical displacement of Q from the x-axis to the distance between the origin and Q.

(C) $\cos\theta = \frac{y}{2}$, because it is the ratio of the vertical displacement of P from the x-axis to the distance between the origin and P.

(D) $\cos\theta = \frac{y}{2}$, because it is the ratio of the vertical displacement of Q from the x-axis to the distance between the origin and Q.

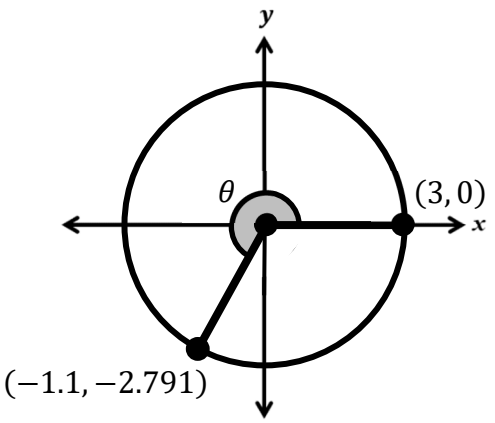
3.2B Sine, Cosine, and Tangent

AP Precalculus Name: _____

CA #1

Each figure below gives a circle in the xy-plane with center at the origin, and an angle θ in standard position. Find the value of each expression.

1.

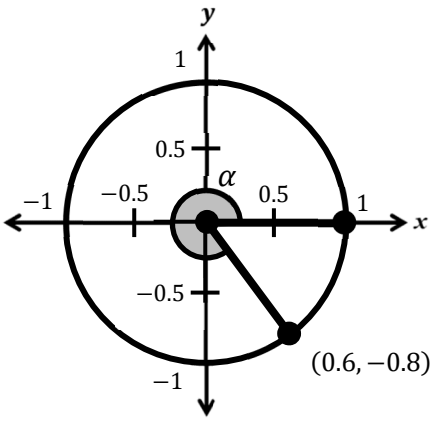


a. $\sin\theta =$

b. $\cos\theta =$

c. $\tan\theta =$

2.

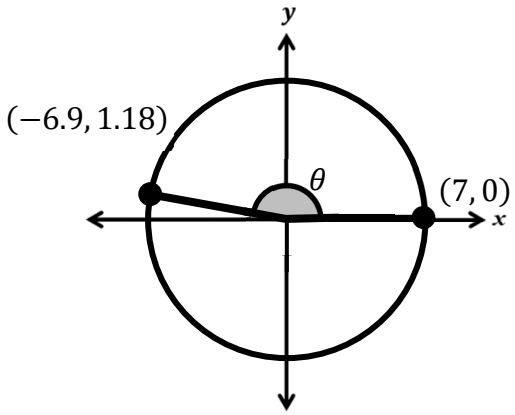


a. $\sin\alpha =$

b. $\cos\alpha =$

c. $\tan\alpha =$

3.

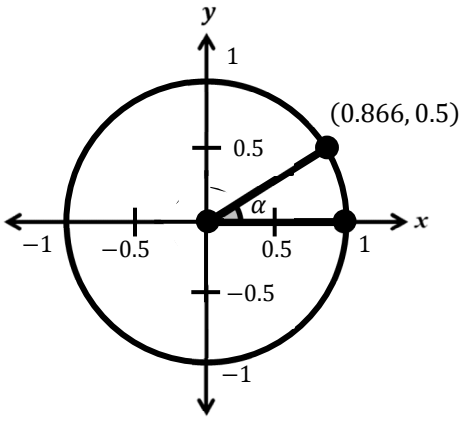


a. $\sin\theta =$

b. $\cos\theta =$

c. $\tan\theta =$

4.



a. $\sin\alpha =$

b. $\cos\alpha =$

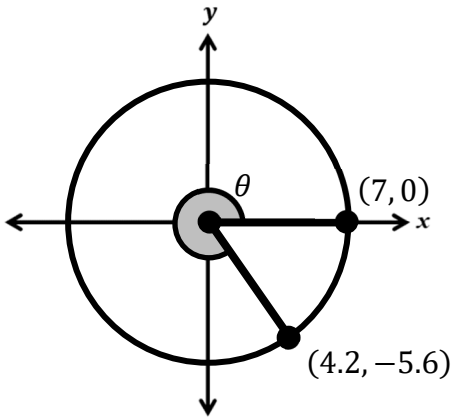
c. $\tan\alpha =$

5.

a. $\sin\theta =$

b. $\cos\theta =$

c. $\tan\theta =$



Answers to 3.2B CA #1

1a. -0.930	2a. -0.8	3a. 0.1685	4a. 0.5	5a. -0.8
1b. -0.3666	2b. 0.6	3b. -0.9857	4b. 0.866	5b. 0.6
1c. 2.537	2c. -1.333	3c. -0.171	4c. 0.577	5c. -1.333

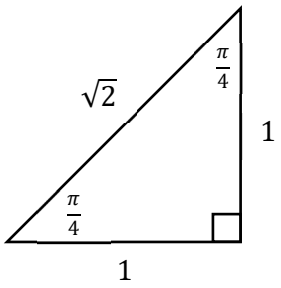
AP Precalc

Write your questions and thoughts here!

3.3A Sine and Cosine Function Values

3.3A Notes

The $45°, 45°, 90°$ triangle in terms of π radians:

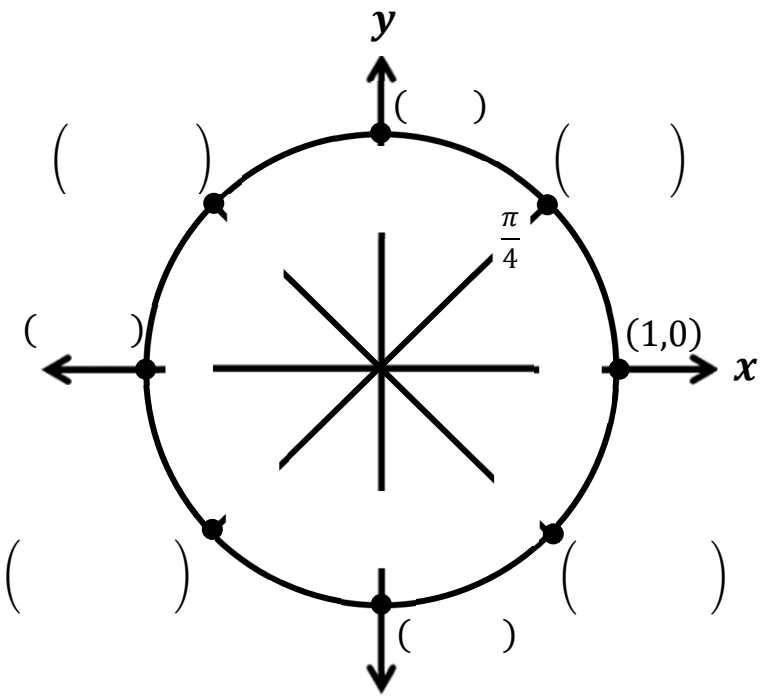


$\mathbf{\sin\frac{\pi}{4}} =$

$\mathbf{\cos\frac{\pi}{4}} =$

The Unit Circle – multiples of $\frac{\pi}{4}$

Recall from our last lesson that on a unit circle $\cos\theta = x$ and $\sin\theta = y$. This allows us to obtain the following coordinate points for all multiples of $\frac{\pi}{4}$.

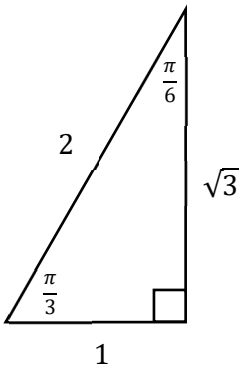


Now we can evaluate sine and cosine for all angles that are multiples of $\frac{\pi}{4}$. While evaluating the below trig expressions, try not to look at the above unit circle. You want to memorize the unit circle so that you can answer these problems quickly. In elementary school, you had multiplication tables that you had to memorize. If you never memorized them, math classes became more challenging. That's how this works as well. If you don't get the coordinate points on the unit circle memorized with their associated angles, then precalculus and calculus become more challenging.

Find the value of each expression. Try not to look back at the Unit Circle for help.					
1. $\sin\frac{\pi}{4}$	2. $\cos\frac{3\pi}{4}$	3. $\sin\pi$	4. $\cos\frac{7\pi}{4}$	5. $\sin\left(-\frac{3\pi}{4}\right)$	6. $\cos\left(-\frac{7\pi}{4}\right)$

Write your questions and thoughts here!

The $30°, 60°, 90°$ triangle in terms of π radians:

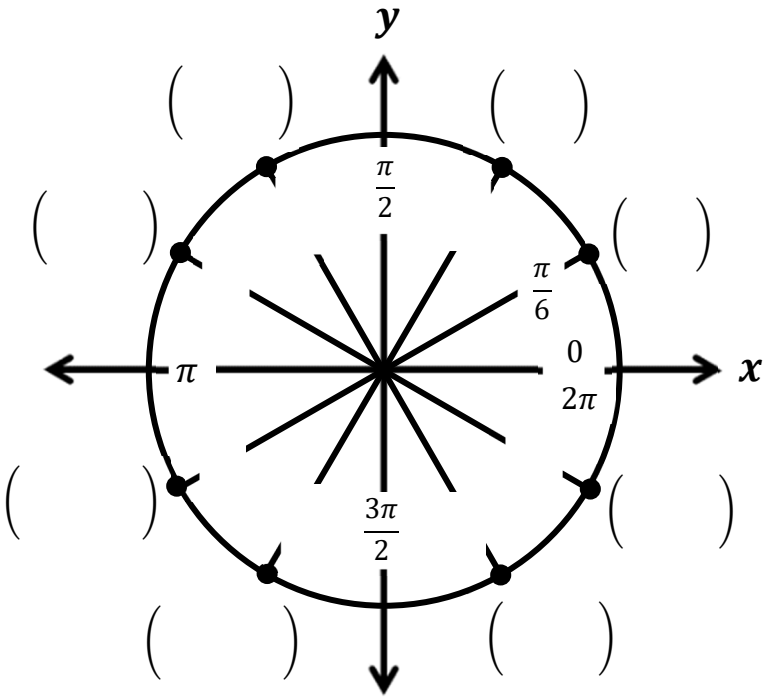


$\mathbf{\sin\frac{\pi}{6}} =$ $\mathbf{\sin\frac{\pi}{3}} =$

$\mathbf{\cos\frac{\pi}{6}} =$ $\mathbf{\cos\frac{\pi}{3}} =$

The Unit Circle – multiples of $\frac{\pi}{6}$

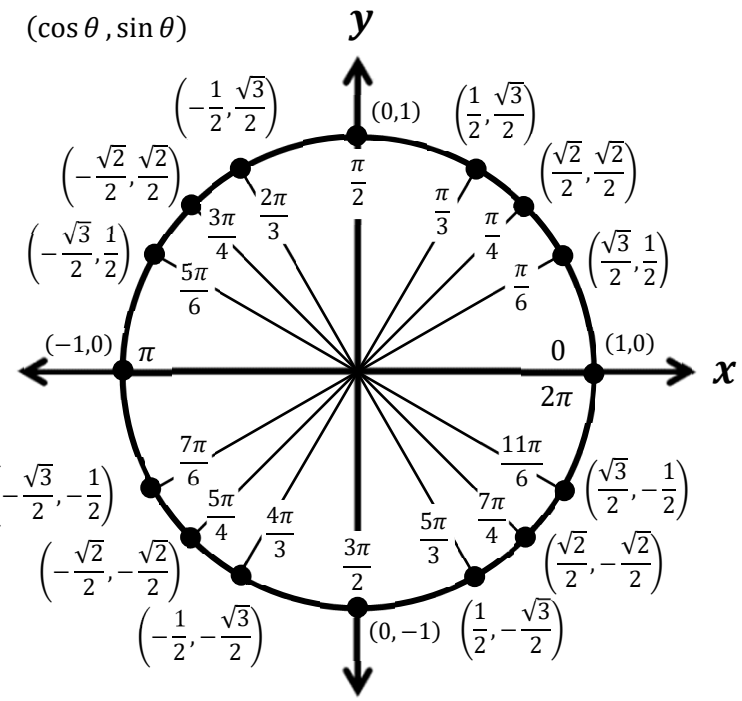
We can now find the coordinate points for all multiples of $\frac{\pi}{6}$.



Find the value of each expression. Try not to look back at the Unit Circle for help.					
7. $\sin\left(-\frac{\pi}{6}\right)$	8. $\cos\frac{\pi}{3}$	9. $\sin\frac{2\pi}{3}$	10. $\cos\left(\frac{5\pi}{6}\right)$	11. $\cos\left(-\frac{\pi}{2}\right)$	12. $\sin\frac{4\pi}{3}$

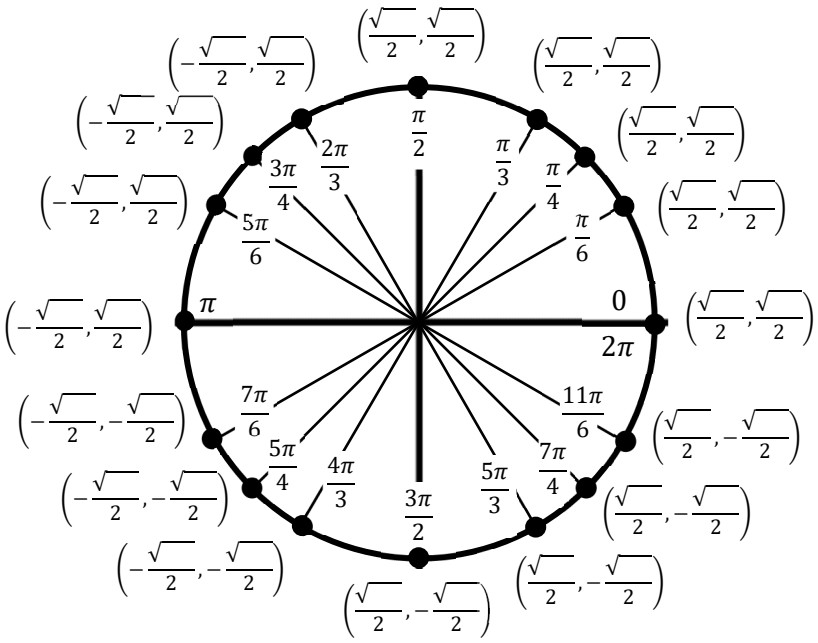
Write your questions and thoughts here!

The completed unit circle:



Tips and Tricks!

- There is a pattern where every coordinate can be written as $\left(\frac{\sqrt{\ }}{2},\frac{\sqrt{\ }}{2}\right)$. Start with "zero" coordinates and count up.
- All the angles that contain $\frac{?}{\mathbf{4}}$ are halfway between the x- and y-axis.
- All the angles that contain $\frac{?}{\mathbf{6}}$ are one angle away from the x-axis.
- All the angles that contain $\frac{?}{\mathbf{3}}$ are one angle away from the y-axis.



3.3A Sine and Cosine Function Values

AP Precalculus

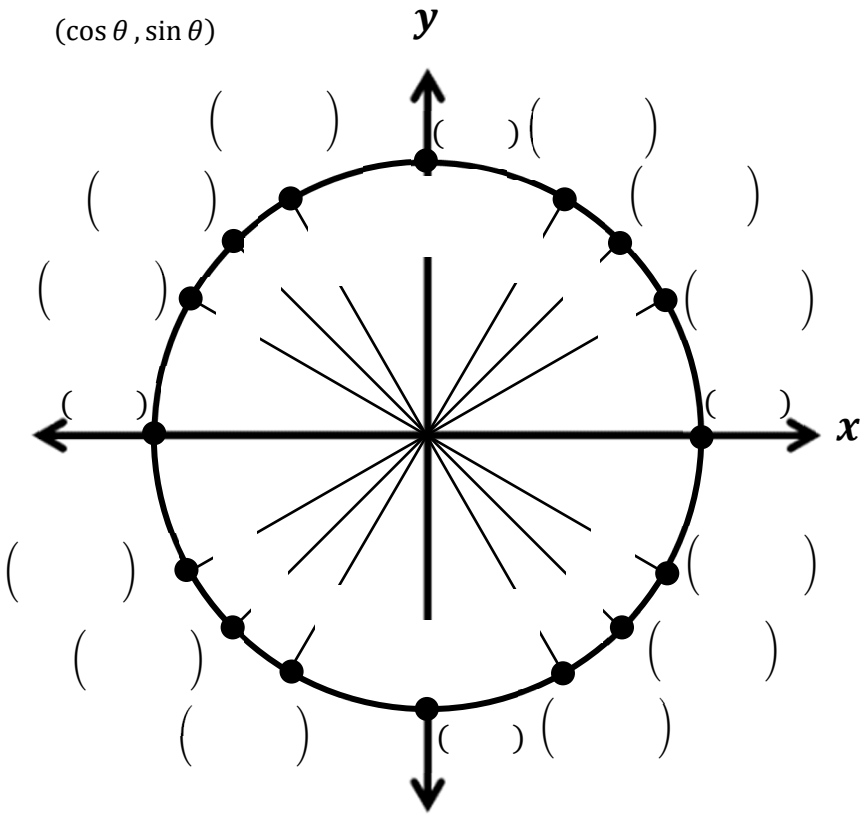
3.3A Practice

Find the value of each expression. Try not to look back at the Unit Circle for help.

1. $\sin\frac{3\pi}{2}$	2. $\cos\frac{\pi}{2}$	3. $\sin\frac{\pi}{6}$	4. $\cos\frac{\pi}{4}$	5. $\sin\frac{5\pi}{6}$
6. $\cos\frac{7\pi}{6}$	7. $\cos\frac{4\pi}{3}$	8. $\sin\frac{5\pi}{3}$	9. $\cos\left(-\frac{7\pi}{6}\right)$	10. $\cos\left(-\frac{\pi}{3}\right)$
11. $\sin\left(-\frac{3\pi}{4}\right)$	12. $\cos\frac{\pi}{3}$	13. $\sin\frac{\pi}{2}$	14. $\sin\frac{2\pi}{3}$	15. $\cos 2\pi$
16. $\cos\frac{5\pi}{6}$	17. $\sin\frac{\pi}{3}$	18. $\cos\left(-\frac{2\pi}{3}\right)$	19. $\sin\frac{\pi}{4}$	20. $\cos\frac{11\pi}{6}$

21. $\sin\left(-\frac{\pi}{6}\right)$	22. $\cos\frac{\pi}{6}$	23. $\sin 0$	24. $\sin\left(-\frac{5\pi}{4}\right)$	25. $\sin\frac{4\pi}{3}$	26. $\sin\pi$

27. Fill in the unit circle below by labeling all of the angles and coordinate points. Do all that you can without looking back at your notes. This will help you know how much you still need to study it.



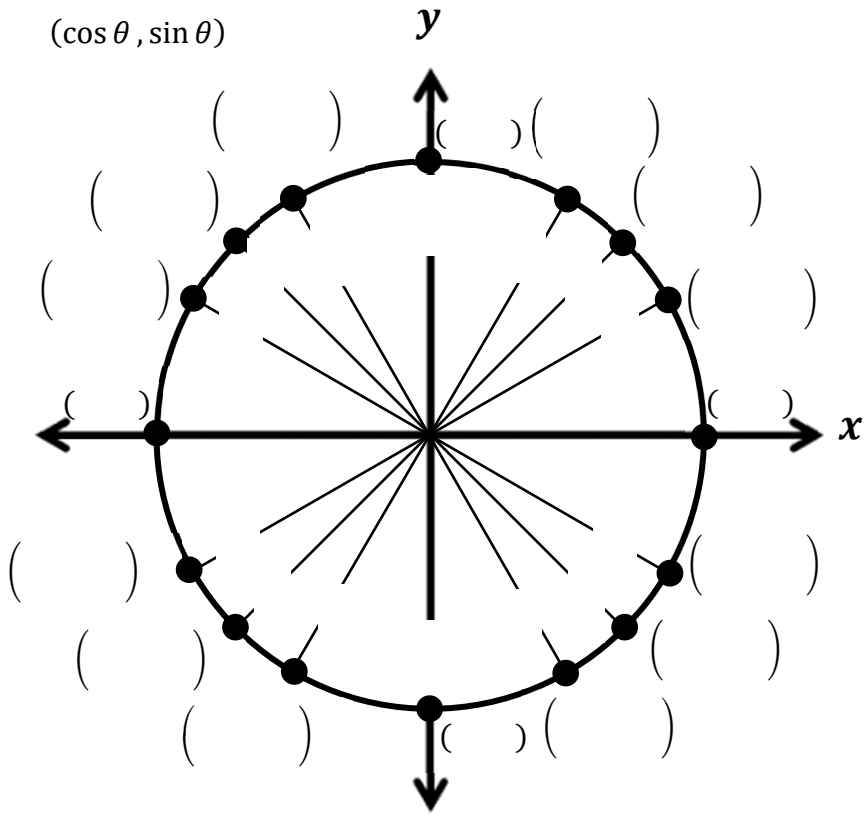
No test prep for this lesson!

3.3A Sine and Cosine Function Values

AP Precalculus Name: _____ **CA #1**

Find the value of each expression. Try not to look back at the Unit Circle for help.				
1. $\cos 2\pi$	2. $\sin\frac{3\pi}{2}$	3. $\cos\frac{\pi}{4}$	4. $\sin\frac{\pi}{3}$	5. $\cos\frac{5\pi}{3}$
6. $\sin\frac{4\pi}{3}$	7. $\cos\frac{5\pi}{6}$	8. $\sin\frac{7\pi}{6}$	9. $\sin\left(-\frac{7\pi}{6}\right)$	10. $\cos\left(-\frac{3\pi}{4}\right)$
11. $\cos\frac{\pi}{3}$	12. $\sin\frac{2\pi}{3}$	13. $\cos\left(-\frac{5\pi}{6}\right)$	14. $\sin\frac{5\pi}{6}$	15. $\cos\frac{\pi}{2}$
16. $\sin\left(-\frac{5\pi}{4}\right)$	17. $\cos\frac{5\pi}{3}$	18. $\sin\frac{\pi}{6}$	19. $\cos\frac{11\pi}{6}$	20. $\sin\pi$

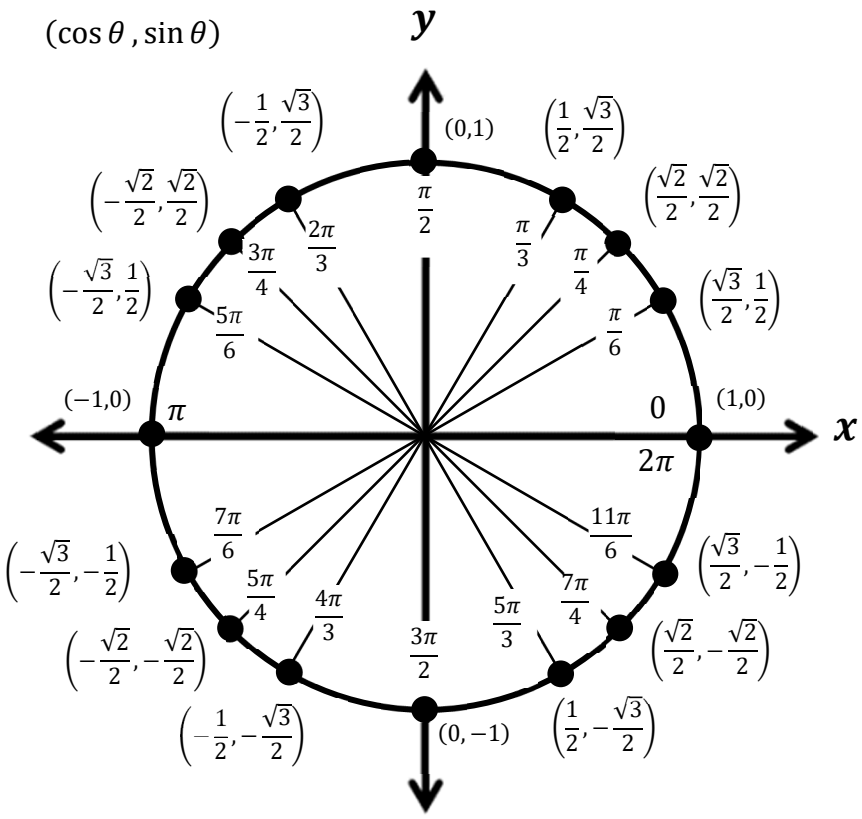
21. Fill in the unit circle below by labeling all of the angles and coordinate points. Do all that you can without looking back at your notes. This will help you know how much you still need to study it.



Answers to 3.3A CA #1

1. 1	2. -1	3. $\frac{\sqrt{2}}{2}$	4. $\frac{\sqrt{3}}{2}$	5. $\frac{1}{2}$	6. $-\frac{\sqrt{3}}{2}$	7. $-\frac{\sqrt{3}}{2}$	8. $-\frac{1}{2}$	9. $\frac{1}{2}$	10. $-\frac{\sqrt{2}}{2}$
11. $\frac{1}{2}$	12. $\frac{\sqrt{3}}{2}$	13. $-\frac{\sqrt{3}}{2}$	14. $\frac{1}{2}$	15. 0	16. $\frac{\sqrt{2}}{2}$	17. $\frac{1}{2}$	18. $\frac{1}{2}$	19. $\frac{\sqrt{3}}{2}$	20. 0

21.



AP Precalc

Write your questions and thoughts here!

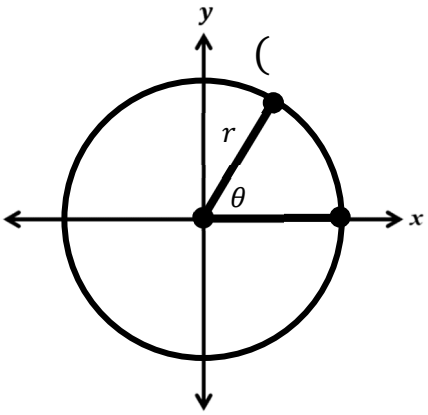
3.3B Sine and Cosine Function Values

3.3B Notes

Recall: $\cos\theta = \frac{x}{r}$ $\sin\theta = \frac{y}{r}$

If we manipulate these equations, we get

$x =$ $y =$



( )

This allows us to find any coordinate point if we know the angle in standard position, and the circle's radius (distance from the origin) where the terminal ray intersects the circle. For this lesson, we will focus on angles that are multiples of $\frac{\pi}{6}$ and $\frac{\pi}{4}$. This means we will not use a calculator, and you will need to have these angles and their coordinate points memorized.

For each problem, an angle in standard position in the xy-plane is given in radians. A circle is centered at the origin with the given radius. What are the coordinates of the point of intersection of the terminal ray of the angle and the circle?

1. $\theta = \frac{5\pi}{6}, r = 6$

2. $\theta = \frac{\pi}{3}, r = 7$

In the xy-plane, the terminal ray of angle θ in standard position intersects a circle of radius r at the given point. What are the values of θ and r?

3. $\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$

4. $\left(-\frac{3\sqrt{3}}{2}, -\frac{3}{2}\right)$

5. $\left(-2, 2\sqrt{3}\right)$

3.3B Sine and Cosine Function Values

AP Precalculus

3.3B Practice

For each problem, an angle in standard position in the xy-plane is given in radians. A circle is centered at the origin with the given radius. What are the coordinates of the point of intersection of the terminal ray of the angle and the circle?

1. $\theta = \frac{\pi}{6}, r = 4$

2. $\theta = \frac{3\pi}{2}, r = 7$

3. $\theta = \frac{7\pi}{6}, r = 6$

4. $\theta = \frac{7\pi}{4}, r = 5$

5. $\theta = \frac{2\pi}{3}, r = 12$

6. $\theta = \frac{5\pi}{3}, r = 3$

7. $\theta = \frac{4\pi}{3}, r = 8$

8. $\theta = \frac{\pi}{3}, r = 16$

9. $\theta = \pi, r = 9$

In the xy-plane, the terminal ray of angle θ in standard position intersects a circle of radius r at the given point. What are the values of θ and r?

10. $(0, -10)$

11. $\left(3\sqrt{2}, 3\sqrt{2}\right)$

12. $\left(-\frac{3\sqrt{2}}{2}, \frac{3\sqrt{2}}{2}\right)$

13. $(-9\sqrt{3}, 9)$

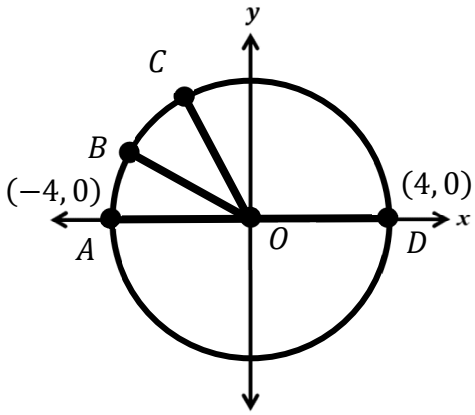
14. $(-15, -15\sqrt{3})$

15. $\left(\frac{7\sqrt{3}}{2}, -\frac{7}{2}\right)$

3.3B Sine and Cosine Function Values

3.3B Test Prep

16. The figure shows a circle of radius 4 along with five labeled points in the xy-plane.



The measure of angle DOC is $\frac{2\pi}{3}$. The measure of angle AOB is half of angle AOC. What are the coordinate points of point B?

(A) $\left(-4\cos\left(\frac{5\pi}{6}\right), 4\sin\left(\frac{5\pi}{6}\right)\right)$

(B) $\left(4\cos\left(\frac{5\pi}{6}\right), 4\sin\left(\frac{5\pi}{6}\right)\right)$

(C) $\left(-2\cos\left(\frac{5\pi}{6}\right), 2\sin\left(\frac{5\pi}{6}\right)\right)$

(D) $\left(2\cos\left(\frac{5\pi}{6}\right), 2\sin\left(\frac{5\pi}{6}\right)\right)$

17. Angles A and B are in standard position in the xy-plane. The measure of angle A is $\frac{\pi}{6}$ radians, and the measure of angle B is $\frac{11\pi}{6}$ radians. The terminal rays of both angles intersect a circle centered at the origin with radius 16. What is the distance between these two points of intersection: the circle and terminal ray of angle A, and the circle and terminal ray of angle B?

(A) $16\sin\frac{\pi}{6} - 16\sin\frac{11\pi}{6}$

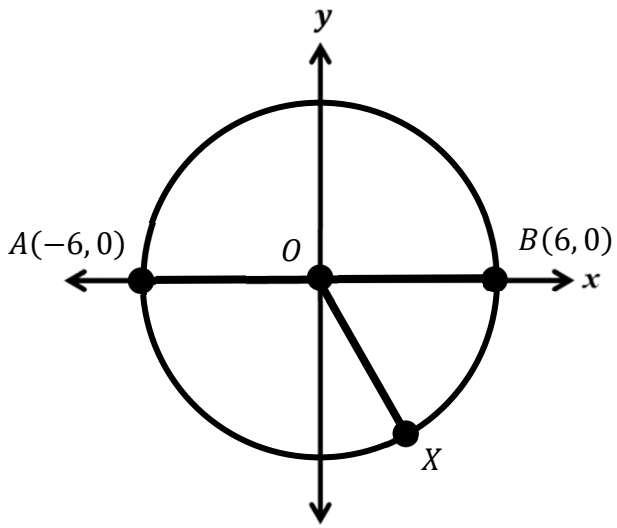
(B) $8\sin\frac{\pi}{6} - 8\sin\frac{11\pi}{6}$

(C) $16\cos\frac{\pi}{6} - 16\cos\frac{11\pi}{6}$

(D) $8\cos\frac{\pi}{6} - 8\cos\frac{11\pi}{6}$

---

18. The figure shows a circle of radius 6 along with the origin and three labeled points in the xy-plane. If the coordinates of X are $\left(3, -3\sqrt{3}\right)$, what is the measurement of angle AOX?



(A) $\frac{\pi}{3}$ (B) $\frac{2\pi}{3}$ (C) $\frac{4\pi}{3}$ (D) $\frac{5\pi}{3}$

3.3B Sine and Cosine Function Values

AP Precalculus Name: _____ **CA #1**

For each problem, an angle in standard position in the xy-plane is given in radians. A circle is centered at the origin with the given radius. What are the coordinates of the point of intersection of the terminal ray of the angle and the circle?

1. $\theta = \frac{\pi}{6}, r = 8$

2. $\theta = \frac{3\pi}{4}, r = 7$

3. $\theta = \frac{\pi}{3}, r = 12$

4. $\theta = \frac{5\pi}{3}, r = 3$

5. $\theta = \frac{11\pi}{6}, r = 5$

6. $\theta = \frac{2\pi}{3}, r = 22$

In the xy-plane, the terminal ray of angle θ in standard position intersects a circle of radius r at the given point. What are the values of θ and r?

7. $(13, 0)$

8. $\left(-2\sqrt{3}, 2\right)$

9. $\left(9\sqrt{2}, -9\sqrt{2}\right)$

10. $\left(-3\sqrt{2}, -3\sqrt{2}\right)$

Answers to 3.3B CA #1

1. $(4\sqrt{3}, 4)$	2. $\left(-\frac{7\sqrt{2}}{2}, \frac{7\sqrt{2}}{2}\right)$	3. $(6, 6\sqrt{3})$	4. $\left(\frac{3}{2}, -\frac{3\sqrt{3}}{2}\right)$	5. $\left(\frac{5\sqrt{3}}{2}, -\frac{5}{2}\right)$
6. $(-11, 11\sqrt{3})$	7. $\theta = 2\pi, r = 13$	8. $\theta = \frac{5\pi}{6}, r = 4$	9. $\theta = \frac{7\pi}{4}, r = 18$	10. $\theta = \frac{5\pi}{4}, r = 6$

AP Precalc

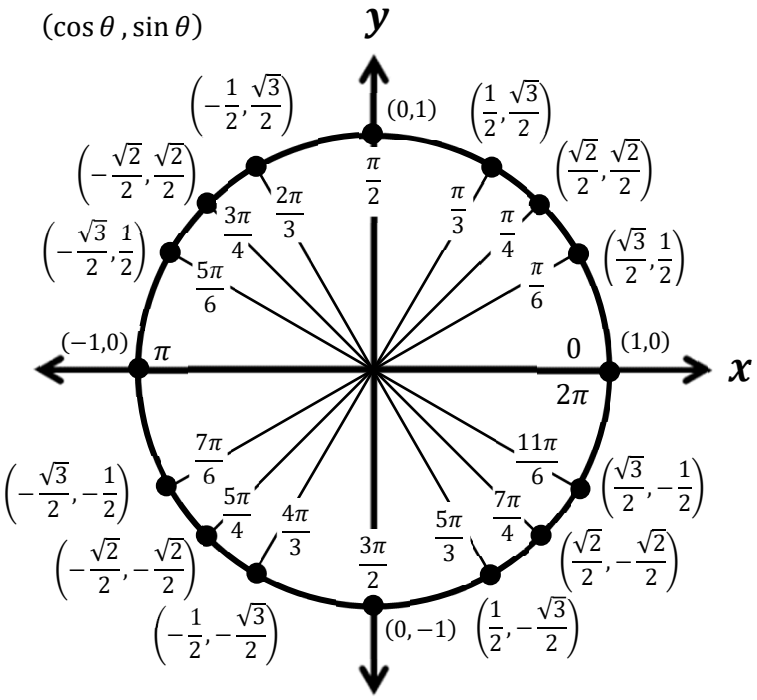
Write your questions and thoughts here!

3.4 Sine and Cosine Function Graphs

3.4 Notes

The Sine Function Graph

Given an angle of measure θ in standard position and a unit circle centered at the origin, there is a point, P, where the terminal ray intersects the circle. The sine function, $f(\theta) = \sin\theta$, gives the y-coordinate, or **vertical displacement** from the ***x*-axis,** of point P.



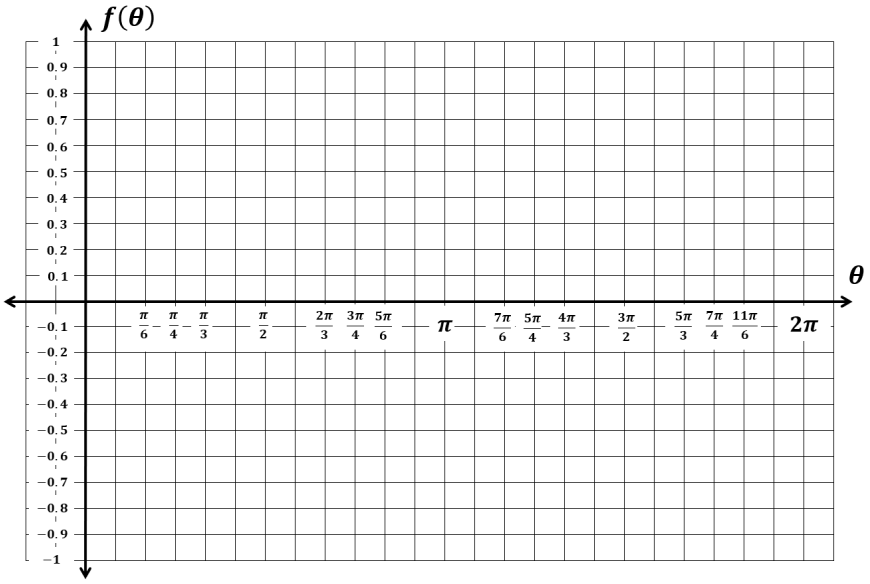
Fill in the table of values for $f(\theta) = \sin\theta$. To save you time, the decimal values have been given.

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π
$\sin\theta$ (exact)									
$\sin\theta$ (decimal)	0	0.5	0.707	0.866	1	0.866	0.707	0.5	0

θ	$\frac{7\pi}{6}$	$\frac{5\pi}{4}$	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	$\frac{7\pi}{4}$	$\frac{11\pi}{6}$	2π
$\sin\theta$ (exact)								
$\sin\theta$ (decimal)	-0.5	-0.707	-0.866	-1	-0.866	-0.707	-0.5	0

Write your questions and thoughts here!

Plot these sine value points and sketch the graph on the axis below.

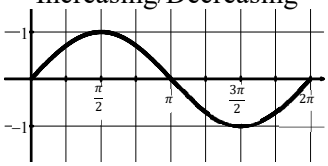


This is considered a periodic function, and you have just graphed one cycle, or period, of the sine function. As we travel around the unit circle the y-values will continue to follow a pattern of increasing to positive one, decreasing to negative one, and then returning to the starting position again of zero.

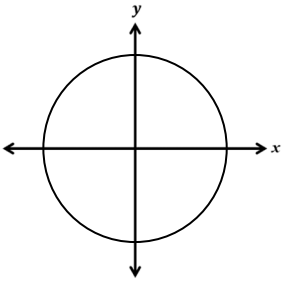
In our next lesson, we will go in depth about how to graph the sine and cosine functions. For this lesson, we will focus on recognizing some basic characteristics of both sine and cosine on the interval $\mathbf{0 \le \theta \le 2\pi}$.

Characteristics of the graph of $f(\theta) = \sin\theta$ on the interval $0 < \theta < 2\pi$.

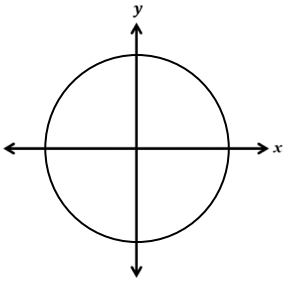
Increasing/Decreasing



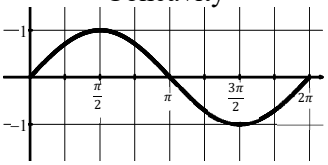
$\sin\theta$ **increases** on



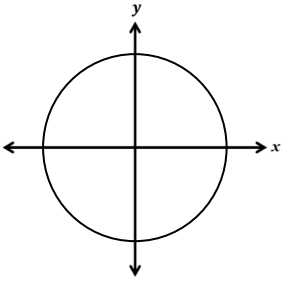
$\sin\theta$ **decreases** on



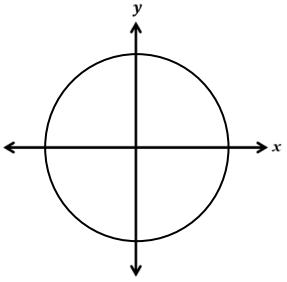
Concavity



$\sin\theta$ is **concave up** on



$\sin\theta$ **concave down** on



Write your questions and thoughts here!

The function f is given by $f(\theta) = \sin\theta$. Describe the concavity of f on the interval, and if f is increasing or decreasing on the interval.

1. $\frac{\pi}{2} < \theta < \pi$	2. $\frac{3\pi}{2} < \theta < 2\pi$	3. $\frac{\pi}{2} < \theta < \frac{3\pi}{2}$

The Cosine Function Graph

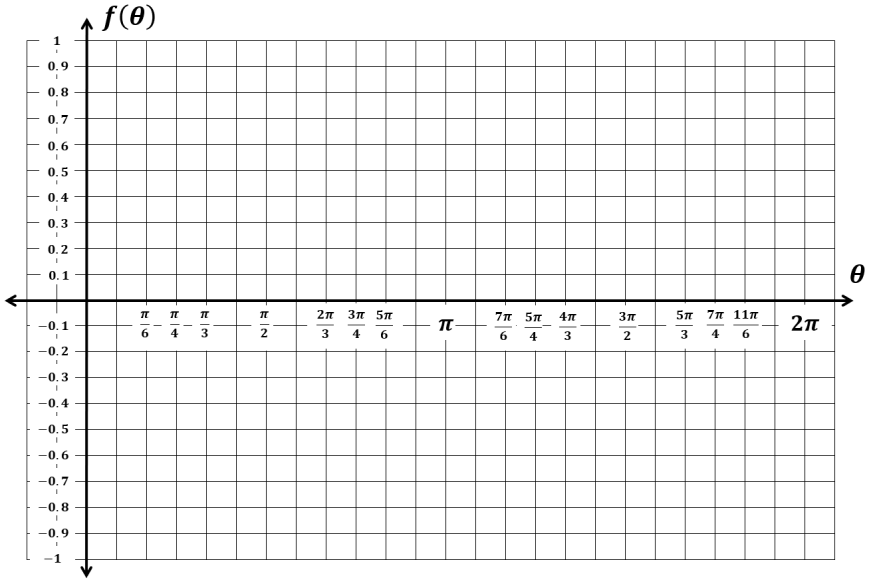
Given an angle of measure θ in standard position and a unit circle centered at the origin, there is a point, P, where the terminal ray intersects the circle. The cosine function, $f(\theta) = \cos\theta$, gives the x-coordinate, or **horizontal displacement** from the **y-axis,** of point P.

Fill in the table of values for $f(\theta) = \cos\theta$. To save you time, the decimal values have been given.

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π
$\cos\theta$ (exact)									
$\cos\theta$ (decimal)	1	0.866	0.707	0.5	0	-0.5	-0.707	-0.866	-1

θ	$\frac{7\pi}{6}$	$\frac{5\pi}{4}$	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	$\frac{7\pi}{4}$	$\frac{11\pi}{6}$	2π
$\cos\theta$ (exact)								
$\cos\theta$ (decimal)	-0.866	-0.707	-0.5	0	0.5	0.707	0.866	1

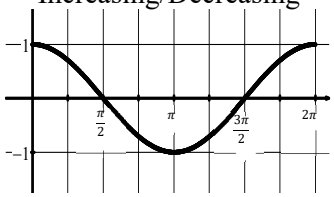
Plot these cosine value points and sketch the graph on the axis below.



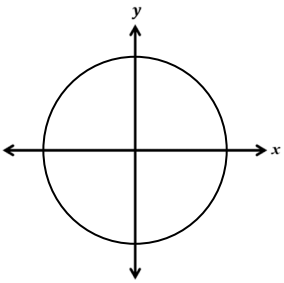
Write your questions and thoughts here!

Characteristics of the graph of $f(\theta) = \cos\theta$ on the interval $0 < \theta < 2\pi$.

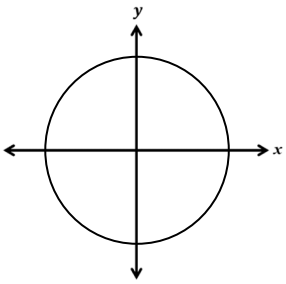
Increasing/Decreasing



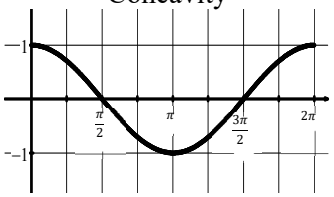
$\cos\theta$ **increases** on



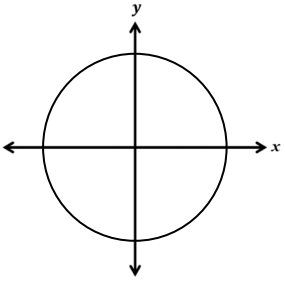
$\cos\theta$ **decreases** on



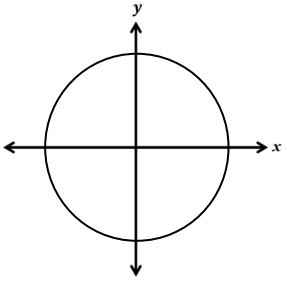
Concavity



$\cos\theta$ is **concave up** on



$\cos\theta$ **concave down** on



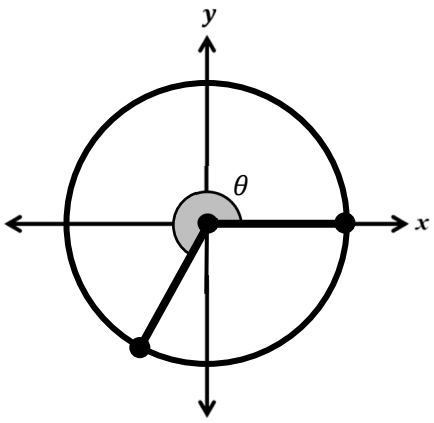
The function f is given by $f(\theta) = \cos\theta$. Describe the concavity of f on the interval, and if f is increasing or decreasing on the interval.

4. $\frac{\pi}{2} < \theta < \pi$

5. $\frac{3\pi}{2} < \theta < 2\pi$

6. $\frac{\pi}{2} < \theta < \frac{3\pi}{2}$

7. We are given an angle θ, in standard position as shown in the figure. The function g is given by $g(a) = \sin a$. For the angle α (not shown), $\theta < \alpha < \frac{3\pi}{2}$. Which of the following is true?



(A) $g(\alpha) < g(\theta)$ (B) $g(\alpha) > g(\theta)$ (C) $g(\alpha) = g(\theta)$

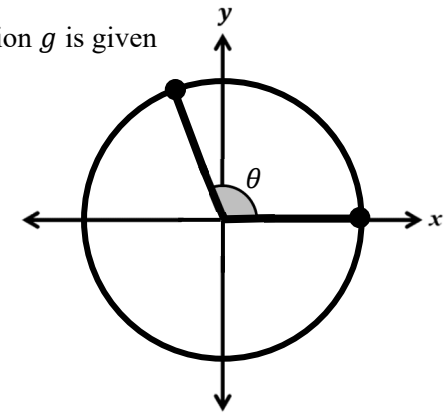
(D) Depending on the value of α, sometimes $g(\alpha) < g(\theta)$ and sometimes $g(\alpha) > g(\theta)$.

3.4 Sine and Cosine Function Graphs

AP Precalculus

3.4 Practice

1. We are given an angle θ, in standard position as shown in the figure. The function g is given by $g(a) = \cos a$. For the angle α (not shown), $\theta < \alpha < \frac{3\pi}{2}$.
Which of the following is true?

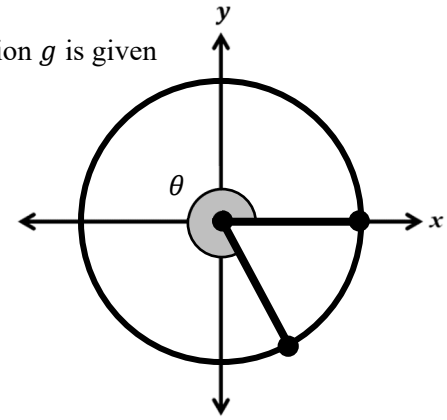


(A) $g(\alpha) < g(\theta)$ (B) $g(\alpha) > g(\theta)$ (C) $g(\alpha) = g(\theta)$

(D) Depending on the value of α, sometimes $g(\alpha) < g(\theta)$ and sometimes $g(\alpha) > g(\theta)$.

---

2. We are given an angle θ, in standard position as shown in the figure. The function g is given by $g(a) = \sin a$. For the angle α (not shown), $\theta < \alpha < 2\pi$.
Which of the following is true?

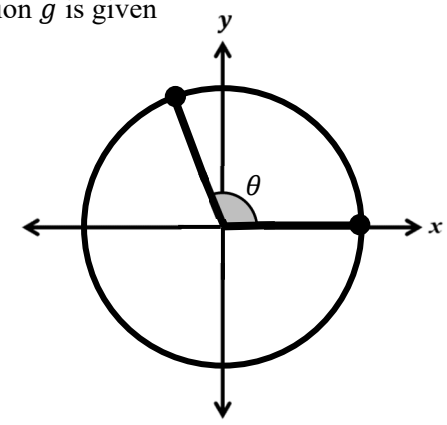


(A) $g(\alpha) < g(\theta)$ (B) $g(\alpha) > g(\theta)$ (C) $g(\alpha) = g(\theta)$

(D) Depending on the value of α, sometimes $g(\alpha) < g(\theta)$ and sometimes $g(\alpha) > g(\theta)$.

---

3. We are given an angle θ, in standard position as shown in the figure. The function g is given by $g(a) = \cos a$. For the angle α (not shown), $\theta < \alpha < 2\pi$.
Which of the following is true?



(A) $g(\alpha) < g(\theta)$ (B) $g(\alpha) > g(\theta)$ (C) $g(\alpha) = g(\theta)$

(D) Depending on the value of α, sometimes $g(\alpha) < g(\theta)$ and sometimes $g(\alpha) > g(\theta)$.

The function f is given by $f(\theta) = \cos\theta$. Describe the concavity of f on the interval, and if f is increasing or decreasing on the interval.

4. $0 < \theta < \frac{\pi}{2}$

5. $\frac{\pi}{2} < \theta < \pi$

6. $\pi < \theta < \frac{3\pi}{2}$

7. $\frac{3\pi}{2} < \theta < 2\pi$

8. $0 < \theta < \pi$

The function f is given by $f(\theta) = \sin\theta$. Describe the concavity of f on the interval, and if f is increasing or decreasing on the interval.

9. $0 < \theta < \frac{\pi}{2}$

10. $\frac{\pi}{2} < \theta < \pi$

11. $\pi < \theta < \frac{3\pi}{2}$

12. $\frac{3\pi}{2} < \theta < 2\pi$

13. $\pi < \theta < 2\pi$

3.4 Sine and Cosine Function Graphs

3.4 Test Prep

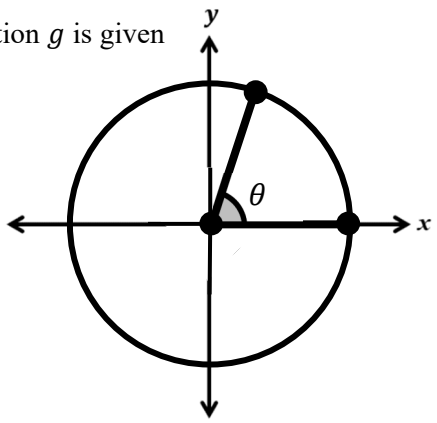
14. For the function $f(\theta) = \cos\theta$, what are all values of the domain when $f(\theta) = 1$?

15. For the function $g(\theta) = \sin\theta$, what are all values of the domain when $g(\theta) = 0$?

3.4 Sine and Cosine Function Graphs

AP Precalculus Name: _____ **CA #1**

1. We are given an angle θ, in standard position as shown in the figure. The function g is given by $g(a) = \sin a$. For the angle α (not shown), $\theta < \alpha < \pi$.
Which of the following is true?

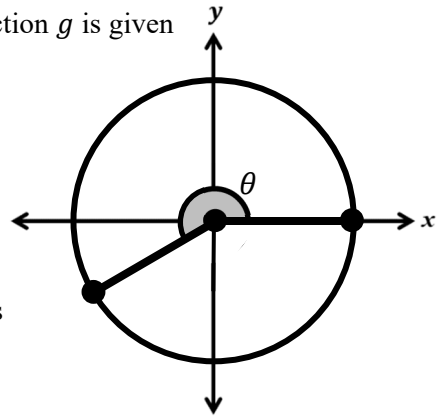


(A) $g(\alpha) < g(\theta)$ (B) $g(\alpha) > g(\theta)$ (C) $g(\alpha) = g(\theta)$

(D) Depending on the value of α, sometimes $g(\alpha) < g(\theta)$ and sometimes $g(\alpha) > g(\theta)$.

---

2. We are given an angle θ, in standard position as shown in the figure. The function g is given by $g(a) = \cos a$. For the angle α (not shown), $\theta < \alpha < \frac{3\pi}{2}$.
Which of the following is true?



(A) $g(\alpha) < g(\theta)$ (B) $g(\alpha) > g(\theta)$ (C) $g(\alpha) = g(\theta)$

(D) Depending on the value of α, sometimes $g(\alpha) < g(\theta)$ and sometimes $g(\alpha) > g(\theta)$.

The function f is given by $f(\theta) = \cos\theta$. Describe the concavity of f on the interval, and if f is increasing or decreasing on the interval.

3. $\frac{\pi}{2} < \theta < \pi$	4. $\pi < \theta < \frac{3\pi}{2}$	5. $\frac{3\pi}{2} < \theta < 2\pi$

The function f is given by $f(\theta) = \sin\theta$. Describe the concavity of f on the interval, and if f is increasing or decreasing on the interval.

6. $0 < \theta < \frac{\pi}{2}$	7. $\frac{\pi}{2} < \theta < \pi$	8. $\frac{3\pi}{2} < \theta < 2\pi$

Answers to 3.4 CA #1

1. D	2. B	3. Concave up Decreasing	4. Concave up Increasing
5. Concave down Increasing	6. Concave down Increasing	7. Concave down Decreasing	8. Concave up Increasing

AP Precalc

Write your questions and thoughts here!

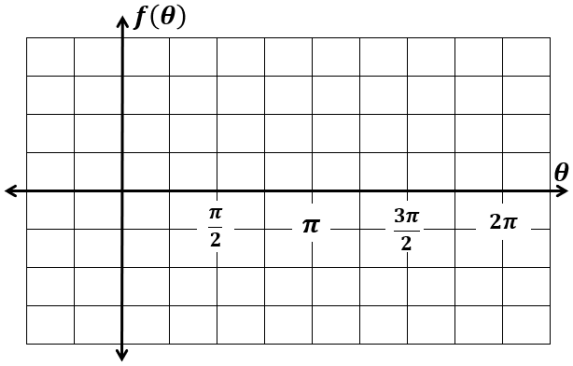
3.5 Sinusoidal Functions

3.5 Notes

Any function that is an additive or multiplicative transformation of $f(\theta) = \sin\theta$ is called a _____ function.

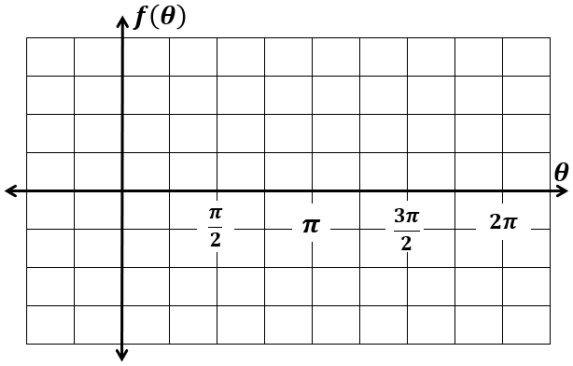
Parent Function Graphs of Sinusoids

$f(\theta) = \cos\theta$



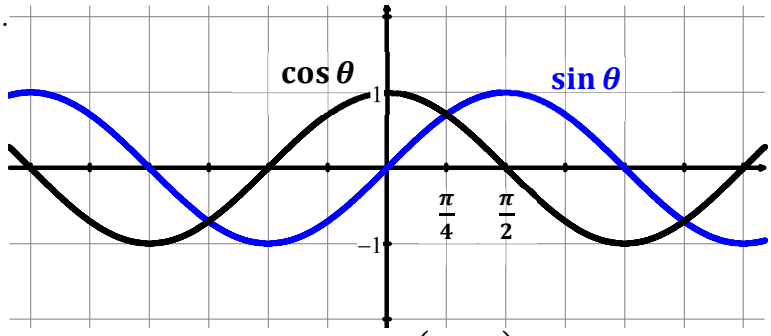
$f(\theta) = \cos\theta$ is an _____ function. It has reflective symmetry over the _____.

$f(\theta) = \sin\theta$



$f(\theta) = \sin\theta$ is an _____ function. It has rotational symmetry about the _____.

$\cos\theta$ is a transformation of $\sin\theta$. If you look at the graphs, we can shift $\sin\theta$ by $\frac{\pi}{2}$ to the left, and it will match up with $\cos\theta$.



$\cos\theta = \sin\left(\theta + \frac{\pi}{2}\right)$

Standard equations: $y = a\cos\theta$ $y = a\sin\theta$

Amplitude: ____ Half the difference between the maximum and minimum values.

Midline: A horizontal line halfway between the maximum and minimum values. It is determined by finding the average of the maximum and minimum values. The concavity of the sinusoid will change when it crosses the midline. For $y = a\cos\theta$ and $y = a\sin\theta$ the midline is _____.

Period (cycle): ____ The reciprocal of frequency. The change in θ values required for the function to complete one full cycle.

Frequency: ____ The reciprocal of period. The number of cycles the graph completes per one radian. For these functions the frequency is approximately 0.159 cycles per radian.

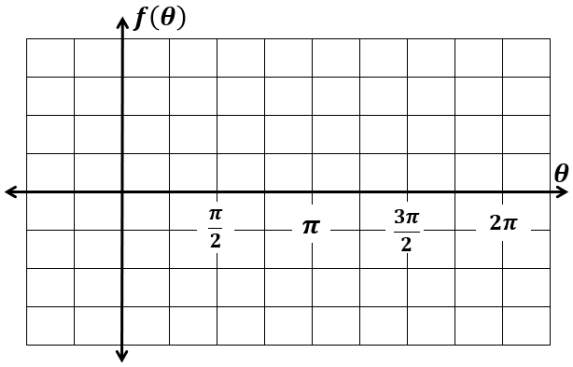
Write your questions and thoughts here!

Graph the following sinusoid functions.

1. $f(\theta) = 3\sin\theta$

Amp: _____ x-ints: _____

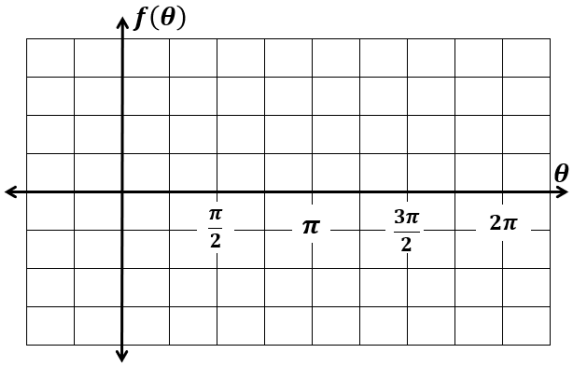
Max value: _____ Min value: _____



2. $f(\theta) = 2\cos(\theta)$

Amp: _____ x-ints: _____

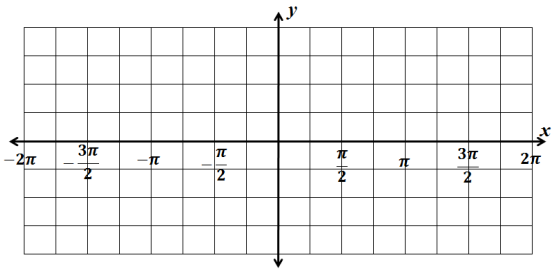
Max value: _____ Min value: _____



3. $y = -2\sin x$

Amp: _____ x-ints: _____

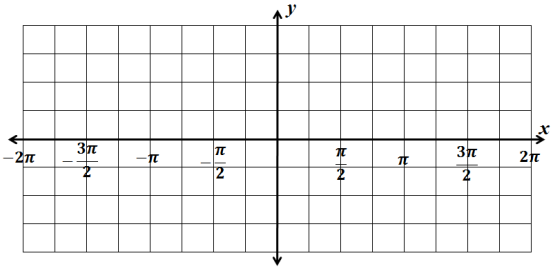
Max value: _____ Min value: _____



4. $y = -4\cos x$

Amp: _____ x-ints: _____

Max value: _____ Min value: _____



For each problem, the sinusoid has been vertically shifted and has the given maximum and minimum values. Write the equation of the midline for the sinusoid AND find the amplitude.

5. Maximum value: 100
   Minimum value: 20

6. Maximum value: 135
   Minimum value: 90

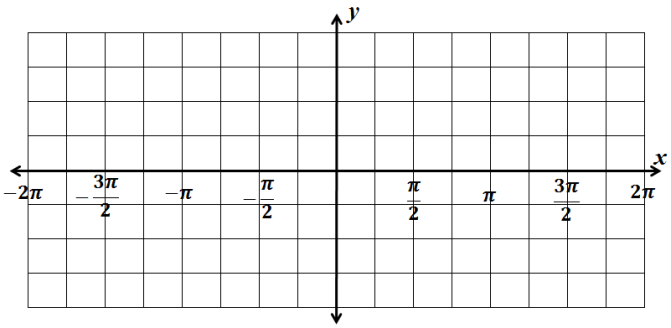
3.5 Sinusoidal Functions

AP Precalculus

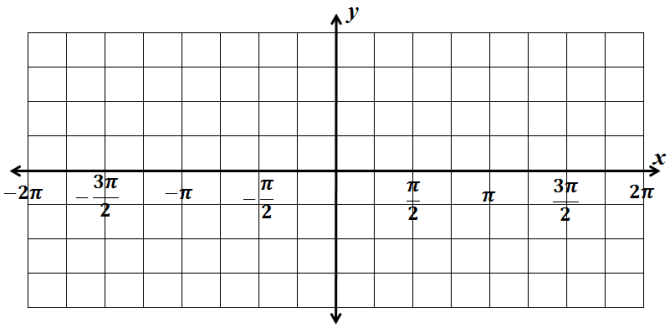
3.5 Practice

Graph the trig function.

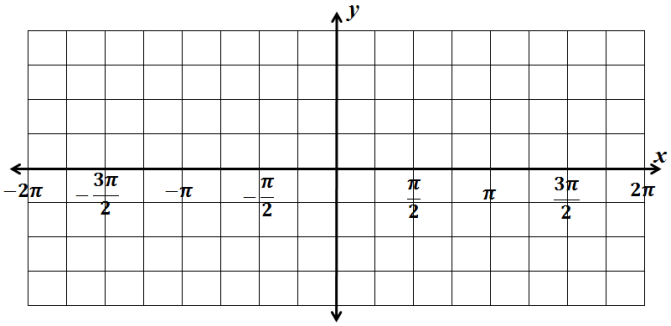
1. $f(\theta) = 3\sin\theta$



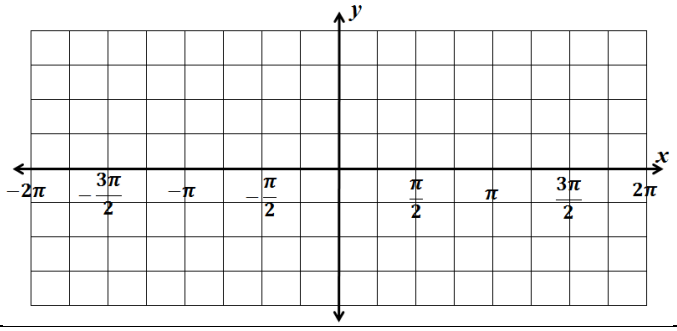
2. $y = -\sin x$



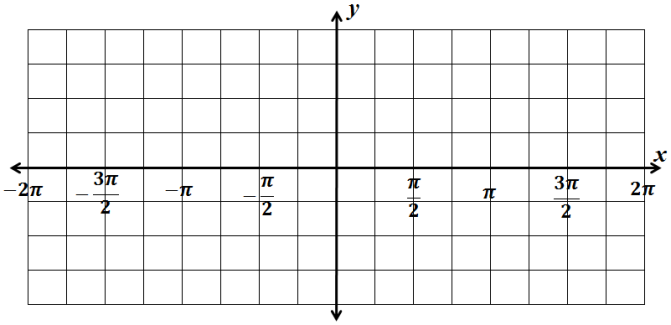
3. $f(\theta) = -2\cos\theta$



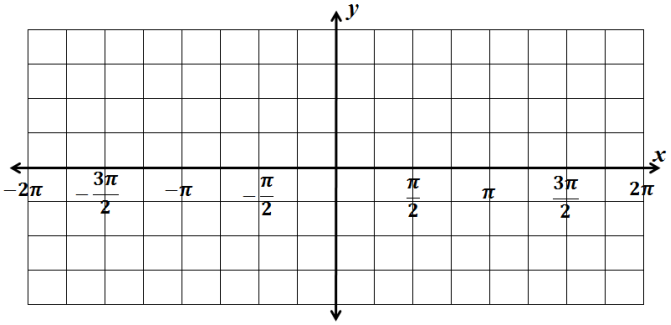
4. $y = -\cos x$



5. $f(\theta) = -2\sin\theta$

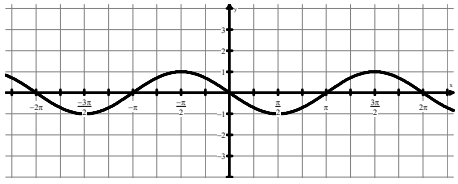


6. $f(t) = 4\cos t$



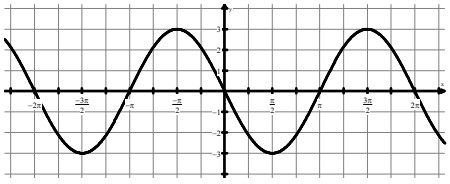
Write the equation of the following *sine* curves.

7.



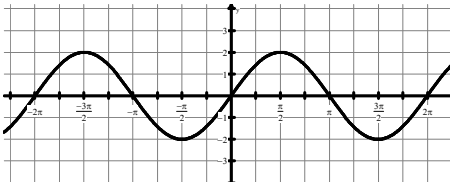
$y =$ _____

8.



$y =$ _____

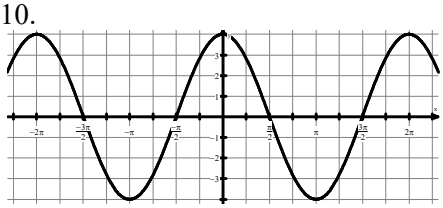
9.



$y =$ _____

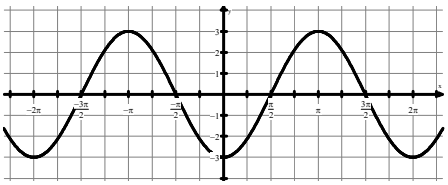
Write the equation of the following *cosine* curves.

10.



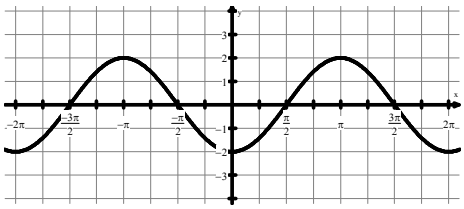
$y =$ _____

11.



$y =$ _____

12.



$y =$ _____

For each problem, the sinusoid has been vertically shifted and has the given maximum and minimum values. Write the equation of the midline for the sinusoid AND find the amplitude.

13. Max value: 40
    Min value: 28

14. Max value: 1.9
    Min value: 0.1

15. Max value: 65
    Min value: 30

16. Max value: 10
    Min value: -6

3.5 Sinusoidal Functions

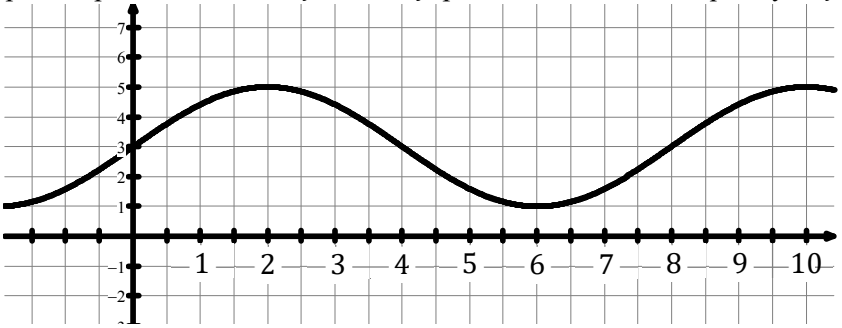
3.5 Test Prep

17. The daily low temperatures of a certain city over a period of time are modeled with a sinusoidal function the xy-plane. The minimum daily low temperature is 24°F, and the maximum daily low temperature is 52°F. Based on these temperatures, which of the following is the best value for the amplitude of the sinusoidal function?

(A) 76 (B) 38 (C) 28 (D) 14

---

18. The figure shows the graph of a periodic function f in the xy-plane. What is the frequency of f?



Graph of f

(A) $\frac{1}{8}$ (B) $\frac{\pi}{8}$ (C) $\frac{\pi}{4}$ (D) 8

19. **Calculator active.** Mr. Brust's patience with his 4th period class seems to have cycles of ups and downs. His patience can be modeled by the function $P(t) = 30\cos(0.15t) + 60$, where t is the number of minutes he has spent with his 4th period class and $P(t)$ is his patience level. A person's patience level is measured as 100 being the most patience anyone can possibly have, and 0 representing no patience. Which of the following best describes the behavior of $P(t)$ on minute 30?
hint: If you are graphing the function on a calculator, be sure your mode is set to RADIANS and not DEGREES.

(A) The amount of patience is increasing at a decreasing rate.

(B) The amount of patience is decreasing at a decreasing rate.

(C) The amount of patience is increasing at an increasing rate.

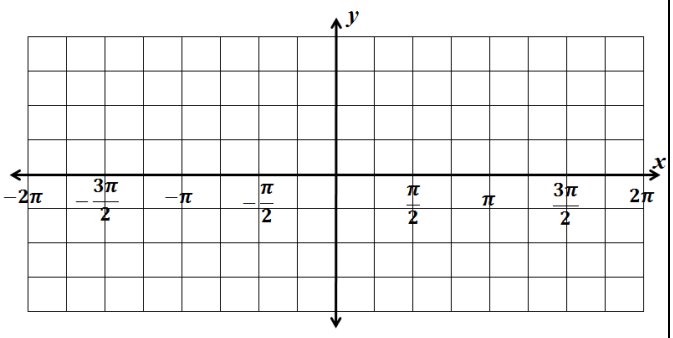
(D) The amount of patience is decreasing at an increasing rate.

3.5 Sinusoidal Functions

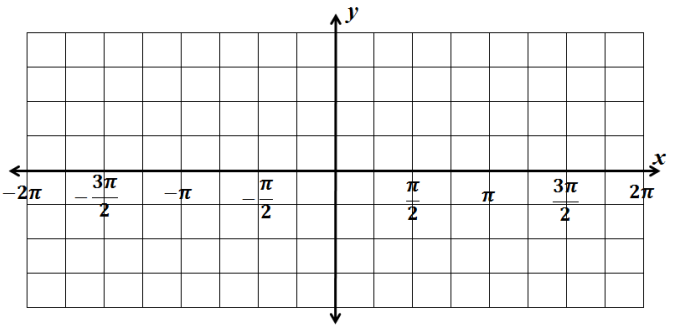
AP Precalculus Name: _____ **CA #1**

Graph the trig function.

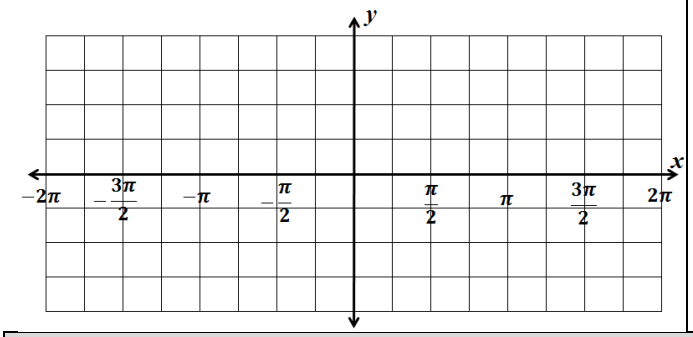
1. $f(\theta) = 3\sin\theta$



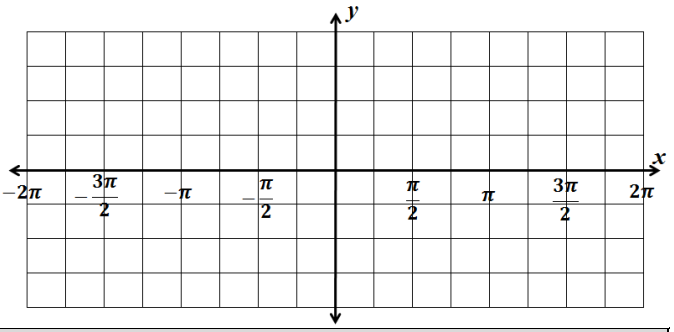
2. $y = -1.5\cos x$



3. $y = -\sin x$

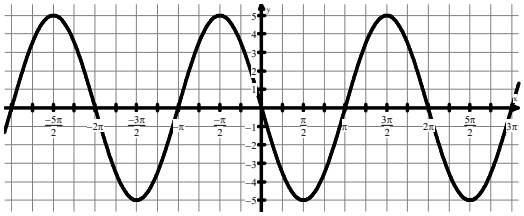


4. $f(\theta) = 3\cos\theta$



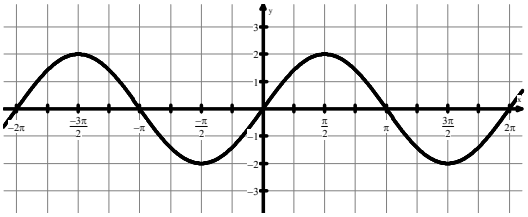
Write the equation of the following *sine* curves.

5.



$y =$ _____

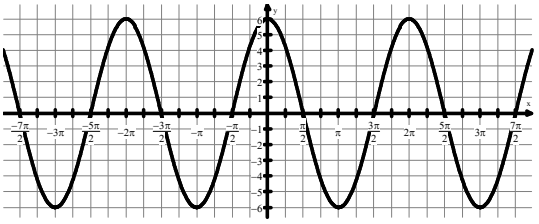
6.



$y =$ _____

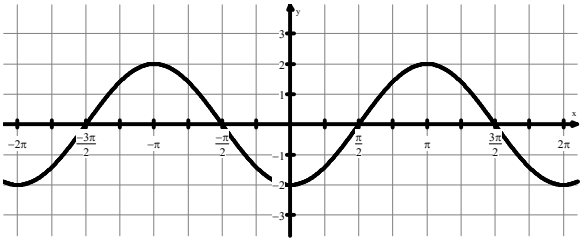
Write the equation of the following *cosine* curves.

7.



$y =$ _____

8.



$y =$ _____

For each problem, the sinusoid has been vertically shifted and has the given maximum and minimum values. Write the equation of the midline for the sinusoid AND find the amplitude.

9.

Max value: 2
Min value: -6

a. What is the equation of the midline?

b. What is the amplitude?

10.

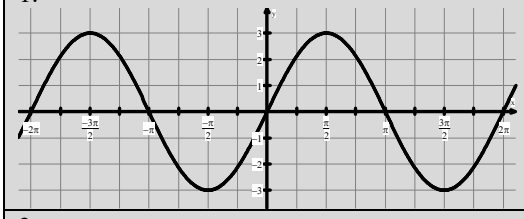
Max value: 13
Min value: 6

a. What is the equation of the midline?

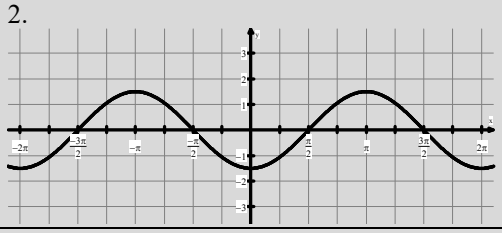
b. What is the amplitude?

Answers to 3.5 CA #1

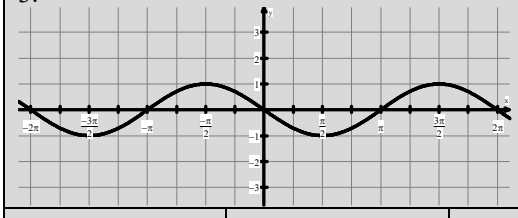
1.



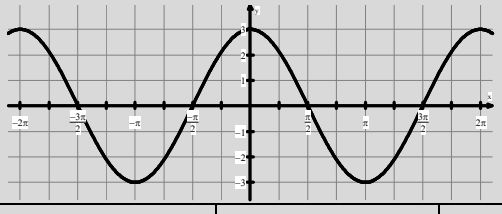
2.



3.



4.



5. $y = -5\sin x$	6. $y = 2\sin x$	7. $y = 6\cos x$	8. $y = -2\cos x$	9a. $y = -2$ 9b. 4	10a. $y = 9.5$ 10b. 3.5

AP Precalc

Write your questions and thoughts here!

3.6A Sinusoidal Function Transformations

3.6A Notes

Standard equations: $\boldsymbol{y = a\cos(b\theta) + d}$ $\boldsymbol{y = a\sin(b\theta) + d}$

Amplitude: ____ Half the difference between the maximum and minimum values.

Midline: A horizontal line halfway between the maximum and minimum values. It is determined by finding the average of the maximum and minimum values. The concavity of the sinusoid will change when it crosses the midline. The equation for the midline is _____.

Period (cycle): ____ The reciprocal of frequency. The change in θ values required for the function to complete one full cycle.

Frequency: ____ The reciprocal of period. The number of cycles the graph completes per one radian.

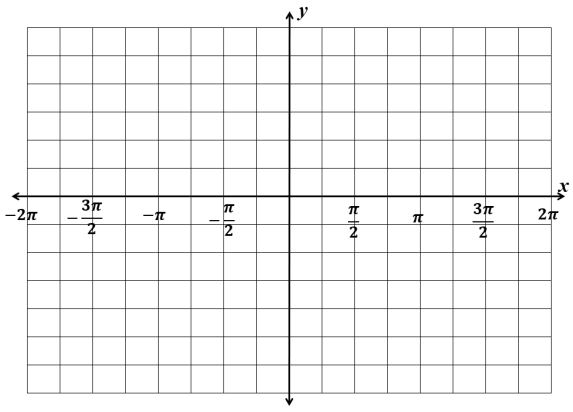
Graph the following sinusoid functions.

1. $f(x) = 3\sin(2x)$

Amp: _____ Period: _____

Midline: _____ Freq: _____

Max value: _____ Min value: _____

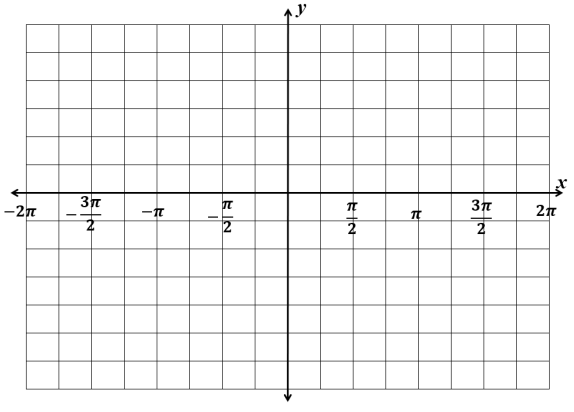


2. $y = -2\cos\left(\frac{x}{2}\right)$

Amp: _____ Period: _____

Midline: _____ Freq: _____

Max value: _____ Min value: _____



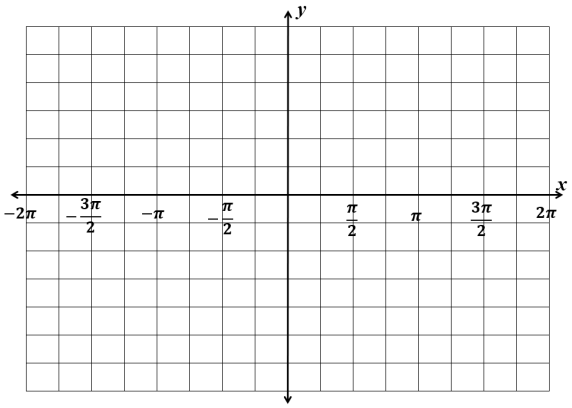
Write your questions and thoughts here!

3. $y = 2\sin\left(\frac{x}{2}\right) + 1$

Amp: _____ Period: _____

Midline: _____ Freq: _____

Max value: _____ Min value: _____

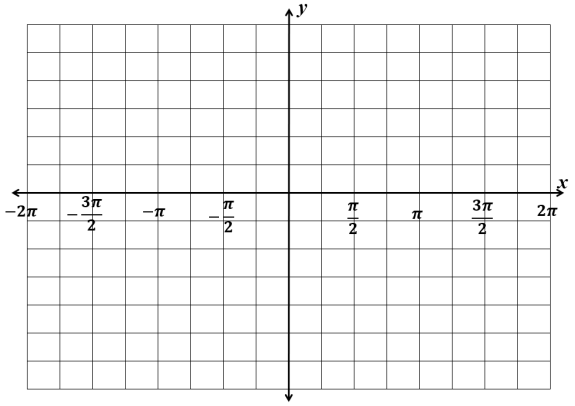


4. $y = 3 - 4\cos(2x)$

Amp: _____ Period: _____

Midline: _____ Freq: _____

Max value: _____ Min value: _____



3.6A Sinusoidal Function Transformations

AP Precalculus

3.6A Practice

Identify the amplitude, period, and frequency of each function.

1. $f(\theta) = 13\sin(2\theta) - 7$

Amp: _____ Period: _____

Freq: _____

2. $g(t) = 5 - 6\cos(\pi t)$

Amp: _____ Period: _____

Freq: _____

3. $h(\theta) = -24\sin(8\pi\theta)$

Amp: _____ Period: _____

Freq: _____

Use the given information to create a *sine* function.

4.

Amplitude: 5
Period: 4π
Vertical Shift: down 4

5.

Amplitude: 2
Period: $\frac{3\pi}{5}$
Vertical Shift: up 9

6.

Amplitude: 1
Period: 4
Vertical Shift: up 1

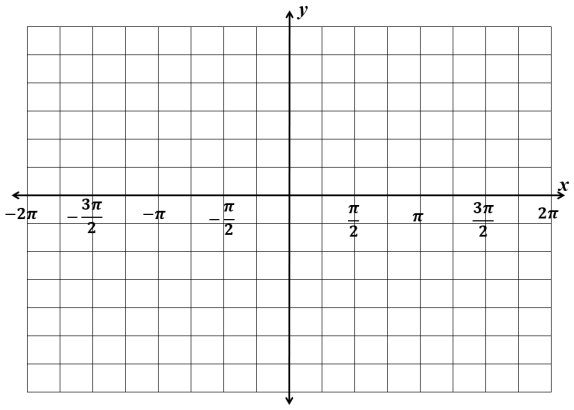
Identify the listed information and graph the trig function.

7. $y = 4\sin\left(\frac{x}{2}\right)$

Amp: _____ Period: _____

Midline: _____ Freq: _____

Max value: _____ Min value: _____

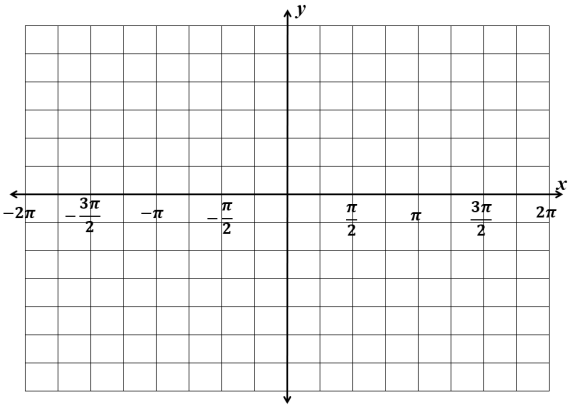


8. $y = -2\cos\frac{1}{2}\theta$

Amp: _____ Period: _____

Midline: _____ Freq: _____

Max value: _____ Min value: _____

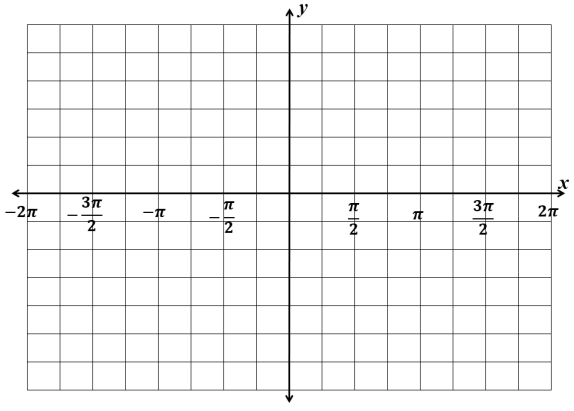


9. $y = -\cos 4x$

Amp: _____ Period: _____

Midline: _____ Freq: _____

Max value: _____ Min value: _____

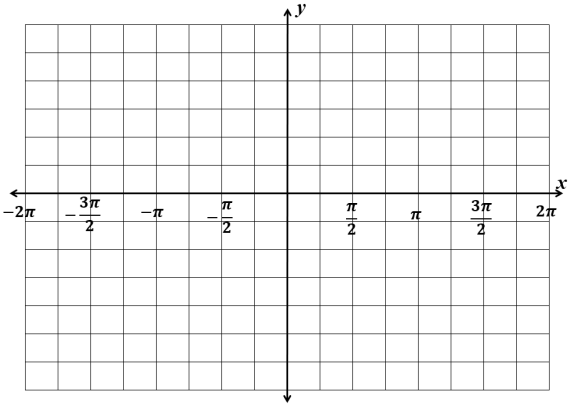


10. $y = -3\cos 2x$

Amp: _____ Period: _____

Midline: _____ Freq: _____

Max value: _____ Min value: _____

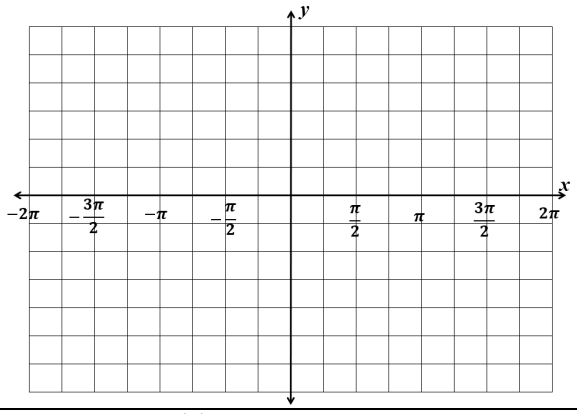


11. $y = 3\sin x - 1$

Amp: _____ Period: _____

Midline: _____ Freq: _____

Max value: _____ Min value: _____

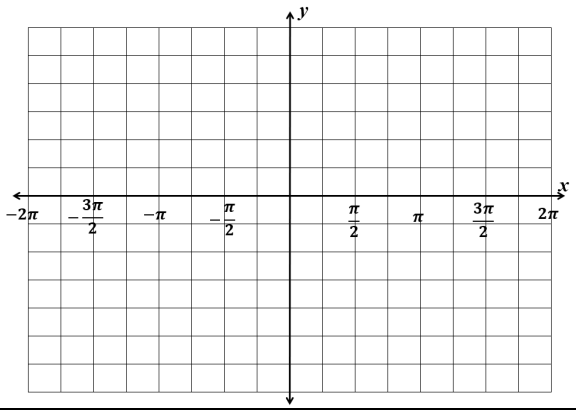


12. $y = -3 + 4\sin\frac{2}{3}x$

Amp: _____ Period: _____

Midline: _____ Freq: _____

Max value: _____ Min value: _____

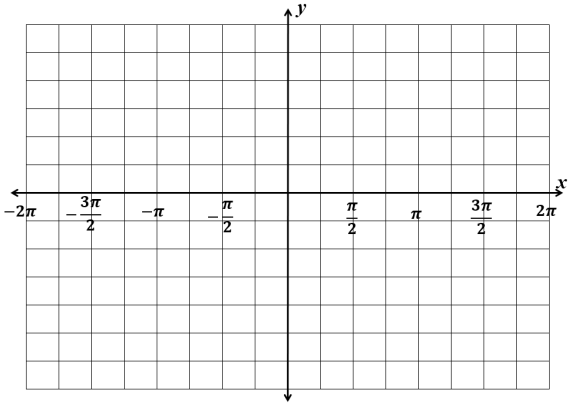


13. $y = 2 - 3\sin\left(\frac{x}{2}\right)$

Amp: _____ Period: _____

Midline: _____ Freq: _____

Max value: _____ Min value: _____

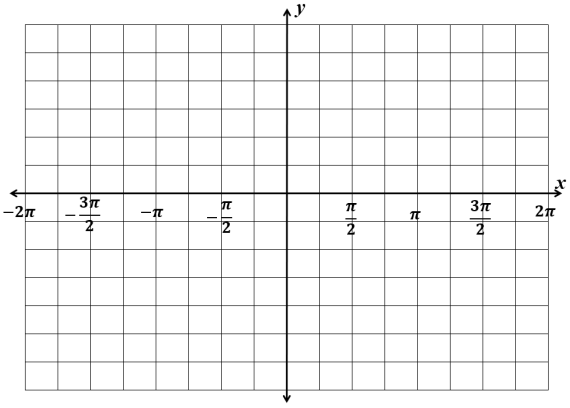


14. $y = -2\cos(2x) + 4$

Amp: _____ Period: _____

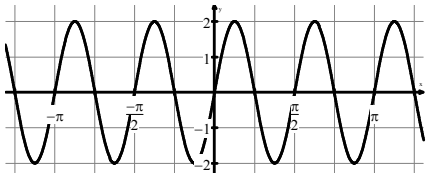
Midline: _____ Freq: _____

Max value: _____ Min value: _____



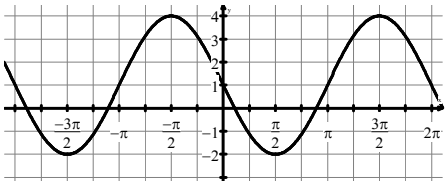
Write the equation of the following *sine* curves.

15.



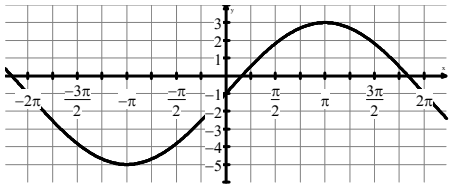
$y =$ _____

16.



$y =$ _____

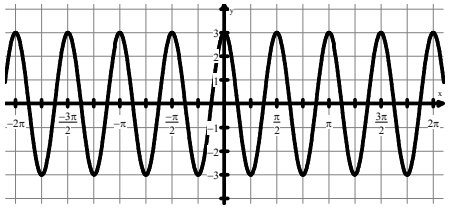
17.



$y =$ _____

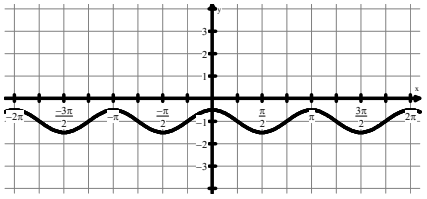
Write the equation of the following *cosine* curves.

18.



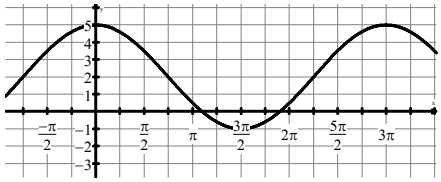
$y =$ _____

19.



$y =$ _____

20.



$y =$ _____

3.6A Sinusoidal Function Transformations

3.6A Test Prep

21. **Calculator active.** The cycle between low tides and high tides at a beachfront town can be measured by

$$f(x) = a\sin(bx) + d$$

where a, b, and d are constants. The maximum value of the tide is 12.04 feet, and the minimum value of the tide is 1.36 feet. Assuming we know the values of b and d, which of the following would best define $f(x)$?

(A) $10.68\sin(bx) + d$

(B) $6.7\sin(bx) + d$

(C) $6.02\sin(bx) + d$

(D) $5.34\sin(bx) + d$

---

22. **Calculator active.** The table gives minimum temperature, in degrees Fahrenheit, on the first day of each of 8 months in a certain city. The function f given by $f(\theta) = a\sin(b\theta) + d$, where a, b, and d are constants, is used to model these data with θ representing the number of the month.

Month	1	2	3	4	5	6	7	8
Temperature (degrees Fahrenheit)	28.6	23.2	34.9	40.3	48.6	56.7	68.2	61.0

Based on the data in the table, which of the following is the best value for d?

(A) $\frac{\pi}{6}$ (B) 22.5 (C) 45 (D) 45.7

23. The table gives ordered pairs for seven points from a larger data set. The larger data set can be modeled by a sinusoidal function f with a period of 6. The minimum values of the data set occur at x-values that are multiples of 6.

x	0	1	2	3	4	5	6
$f(x)$	-4	-1	3	6	3	-1	-4

Which of the following best defines $f(x)$ for the larger data set?

(A) $-4\cos(12\pi x) + 1$

(B) $-4\cos\left(\frac{\pi}{3}x\right) + 1$

(C) $-5\cos(12\pi x) + 1$

(D) $-5\cos\left(\frac{\pi}{3}x\right) + 1$

3.6A Sinusoidal Function Transformations

AP Precalculus Name: _____ **CA #1**

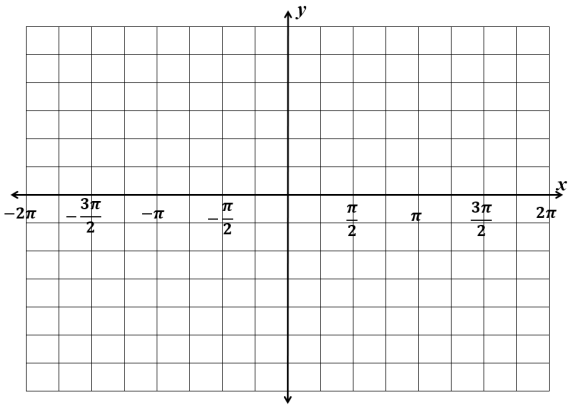
Identify the given information and graph the trig function.

1. $y = -\cos 2x$

Amp: _____ Period: _____

Midline: _____ Freq: _____

Max value: _____ Min value: _____

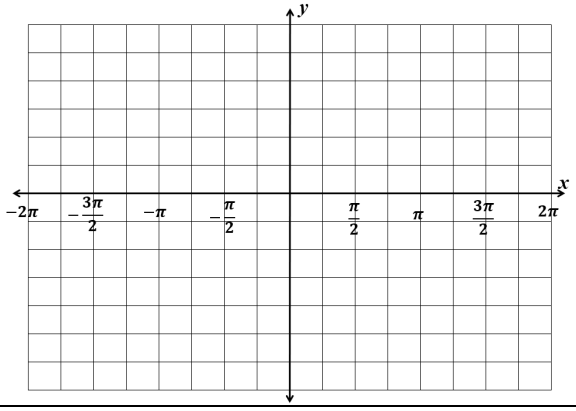


2. $y = 3\sin\frac{1}{2}x$

Amp: _____ Period: _____

Midline: _____ Freq: _____

Max value: _____ Min value: _____

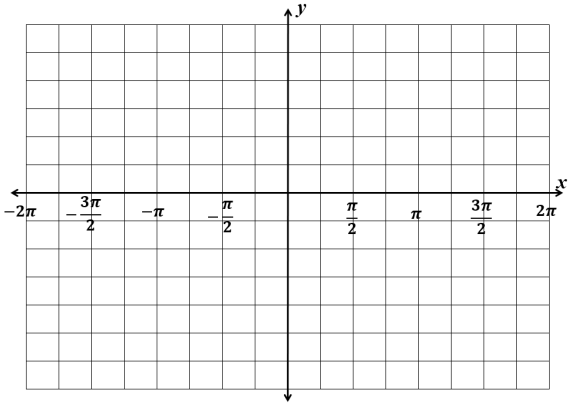


3. $y = 1 - 3\sin 4x$

Amp: _____ Period: _____

Midline: _____ Freq: _____

Max value: _____ Min value: _____

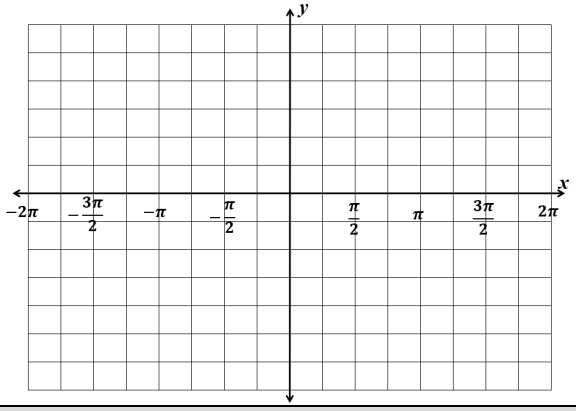


4. $y = 2\cos\frac{1}{3}x - 1$

Amp: _____ Period: _____

Midline: _____ Freq: _____

Max value: _____ Min value: _____



Use the given information to create a *sine* function.

5.

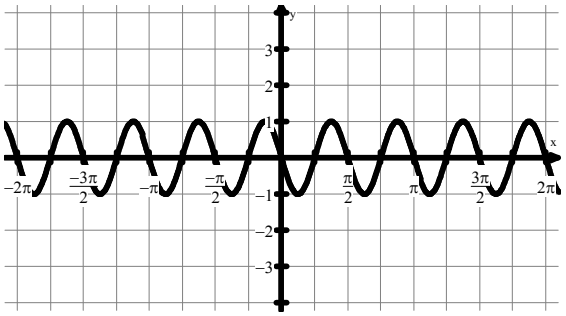
Amplitude: 11
Period: 6π
Vertical Shift: up 3

6.

Amplitude: 6
Period: $\frac{1}{4}$
Vertical Shift: up 7

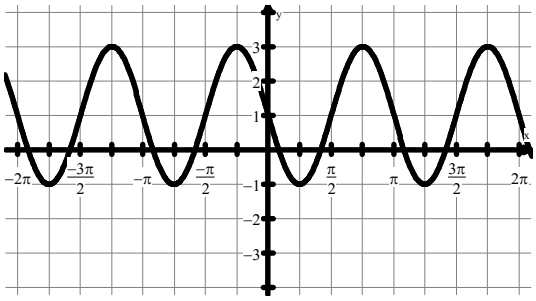
Write the equation of the following *sine* curves.

7.



$y =$ _____

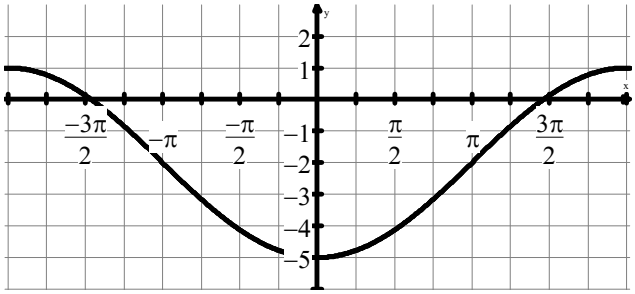
8.



$y =$ _____

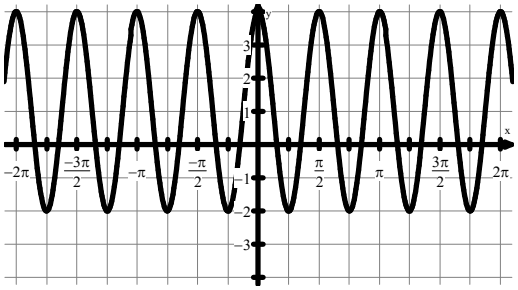
Write the equation of the following *cosine* curves.

9.



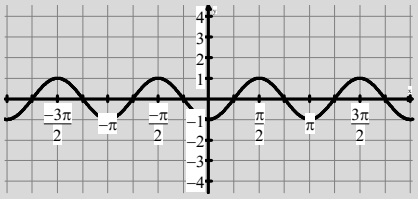
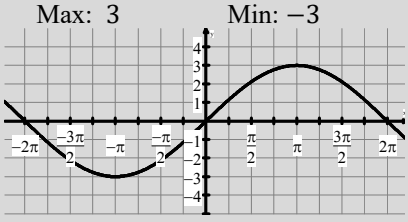
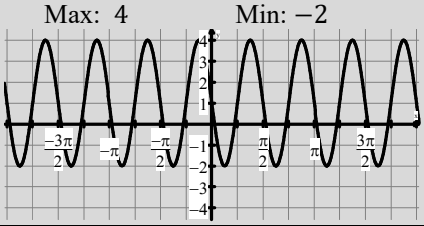
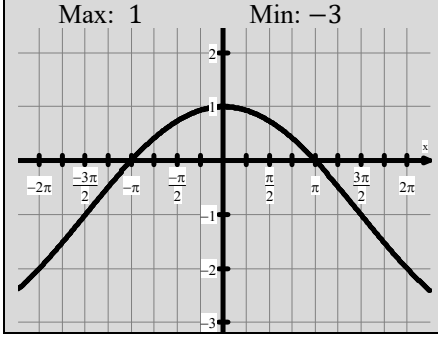
$y =$ _____

10.



$y =$ _____

Answers to 3.6A CA #1

1. Amp: 1 Period: π Midline: $y = 0$ Freq: $\frac{1}{\pi}$ Max: 1 Min: -1 	2. Amp: 3 Period: 4π Midline: $y = 0$ Freq: $\frac{1}{4\pi}$ Max: 3 Min: -3 	3. Amp: 3 Period: $\frac{\pi}{2}$ Midline: $y = 1$ Freq: $\frac{2}{\pi}$ Max: 4 Min: -2 
4. Amp: 2 Period: 6π Midline: $y = -1$ Freq: $\frac{1}{6\pi}$ Max: 1 Min: -3 	5. $y = 11\sin\left(\frac{1}{3}x\right) + 3$	6. $y = 6\sin(8\pi x) + 7$
	7. $y = -\sin 4x$	8. $y = 1 - 2\sin(2x)$
	9. $y = -3\cos\left(\frac{x}{2}\right) - 2$	10. $y = 3\cos(4x) + 1$

AP Precalc

Write your questions and thoughts here!

3.6B Sinusoidal Function Transformations

3.6B Notes

The focus of this lesson is to learn about the phase shift of a periodic function. This means we are shifting the sinusoidal function horizontally (to the left or right).

Standard equations:

$$y = a\cos\big(b(\theta \quad)\big) + d \qquad y = a\sin\big(b(\theta \quad)\big) + d$$

Graph the following sinusoid functions.

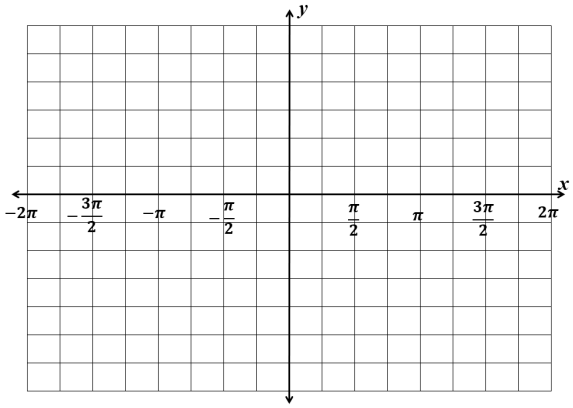
1. $f(x) = 3\cos\left(x + \frac{\pi}{4}\right) + 2$

Amp: _____ Period: _____

Midline: _____ Freq: _____

Max value: _____ Min value: _____

Phase Shift: _____



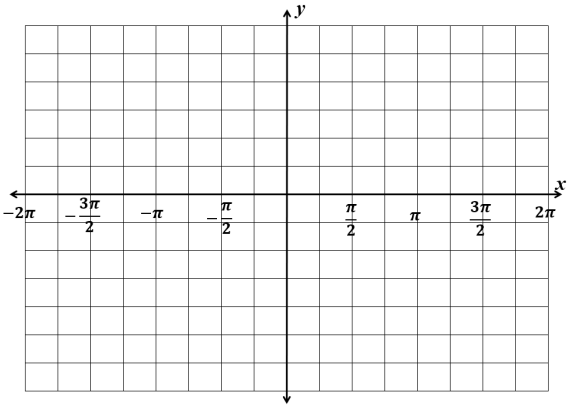
2. $y = -2\sin(2x - \pi) - 1$

Amp: _____ Period: _____

Midline: _____ Freq: _____

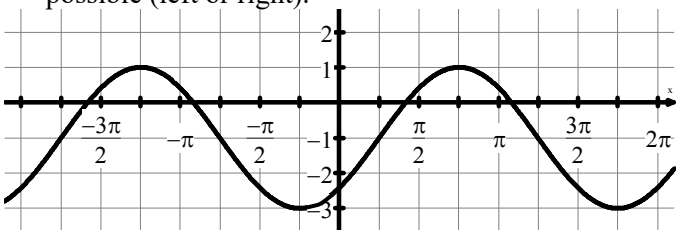
Max value: _____ Min value: _____

Phase Shift: _____



3. Create a *sine* function that has the following attributes.
   Amplitude: 3
   Period: π
   Phase Shift: left $\frac{\pi}{8}$
   Vertical Shift: down 4

4. Write the equation of the following *sine* curve. Use a positive leading coefficient and the closest phase shift possible (left or right).



3.6B Sinusoidal Functions Transformations

AP Precalculus

3.6B Practice

Identify the following attributes of each function.

1. $f(x) = 13\sin\left(4\left(x + \frac{\pi}{11}\right)\right) - 7$

Amp: _____ Period: _____

Phase Shift: _____

Vertical Shift: _____

2. $f(\theta) = 2\sin\left(\frac{\theta}{2} - \frac{\pi}{5}\right) - 5$

Amp: _____ Period: _____

Phase Shift: _____

Vertical Shift: _____

3. $f(x) = 6\cos\left(3x + \frac{\pi}{6}\right) + 11$

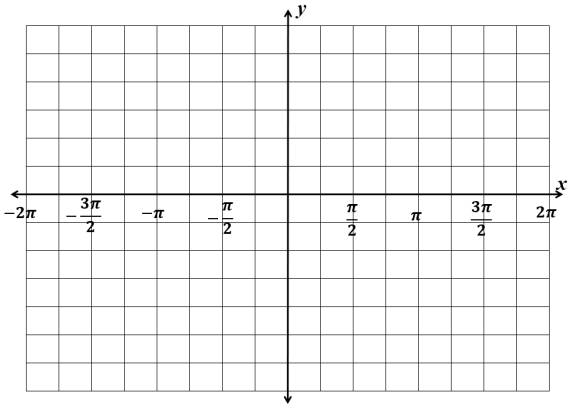
Amp: _____ Period: _____

Phase Shift: _____

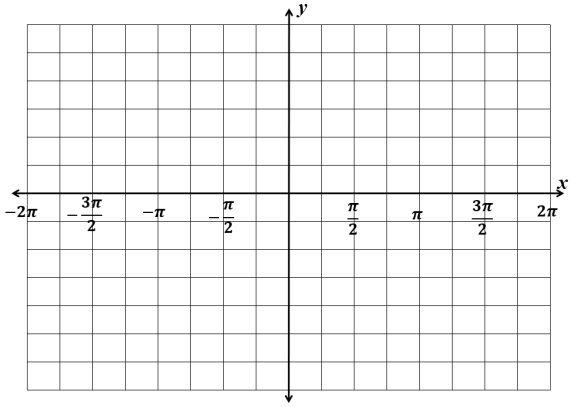
Vertical Shift: _____

Graph the trig function.

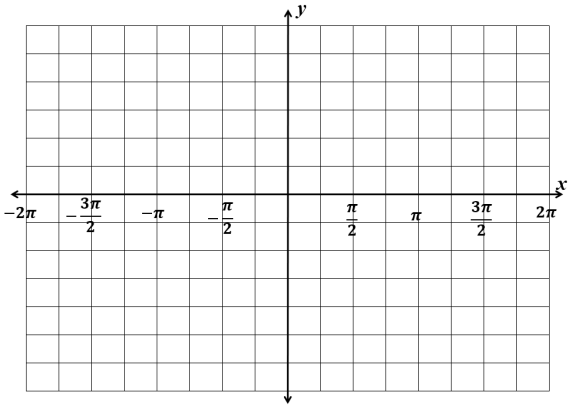
4. $y = 2\sin\left(x + \frac{\pi}{4}\right)$



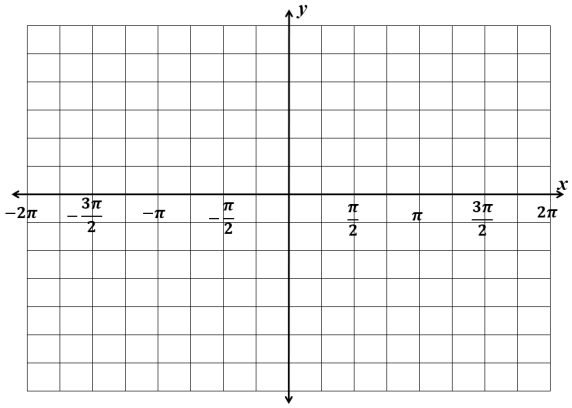
5. $y = 2 + \sin\left(2\left(x + \frac{\pi}{4}\right)\right)$



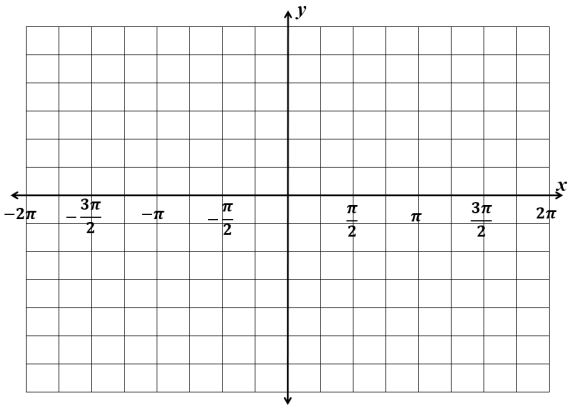
6. $y = -4\cos\left(\frac{1}{2}\left(x - \frac{\pi}{2}\right)\right) - 2$



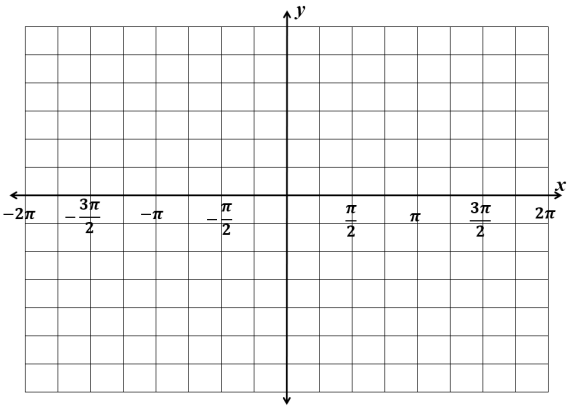
7. $y = -2\sin(2x + \pi) + 1$



8. $y = \cos\left(2x - \frac{\pi}{2}\right)$



9. $y = -3\sin\left(\frac{x}{2} + \frac{\pi}{2}\right) + 1$



Create a sine function that matches the following attributes.

10.
Amplitude: 2
Period: $\frac{3\pi}{2}$
Phase Shift: left $\frac{5\pi}{9}$
Vertical Shift: down 14

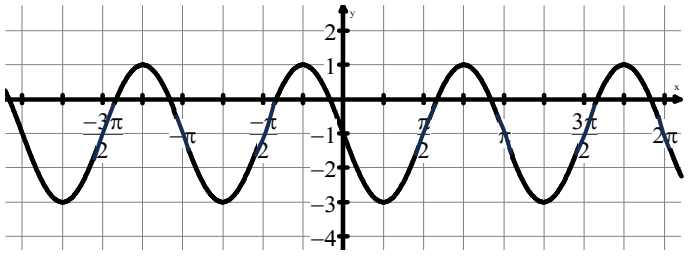
11.
Amplitude: 5
Period: $\frac{\pi}{6}$
Phase Shift: right $\frac{\pi}{24}$
Vertical Shift: up 8

12.
Amplitude: 1
Period: 6
Phase Shift: left $\frac{6\pi}{7}$
Vertical Shift: up 2

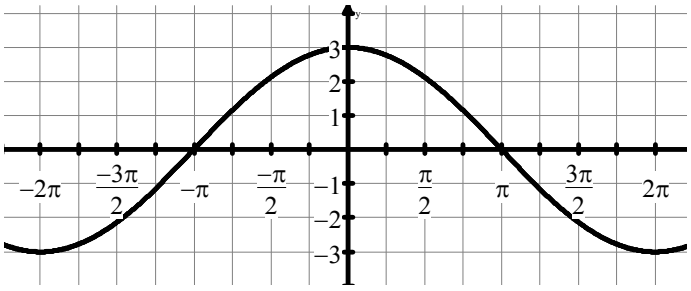
13.
Amplitude: 8
Period: $\frac{2\pi}{3}$
Phase Shift: right $\frac{\pi}{21}$
Vertical Shift: up 5

Write a sine AND cosine function for the following curves. Use a positive leading coefficient and the closest phase shift possible (left or right). For some problems, it may be equal to move left or right.

14.



15.



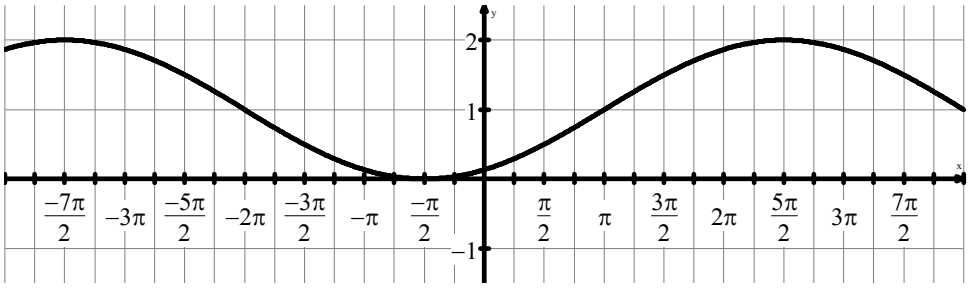
Sine equation: $y =$ _____

Cosine equation: $y =$ _____

Sine equation: $y =$ _____

Cosine equation: $y =$ _____

16.



Sine equation: $y =$ _____

Cosine equation: $y =$ _____

3.6B Sinusoidal Functions Transformations

3.6B Test Prep

17. Suppose you are riding a Ferris wheel. After everyone is loaded, the wheel starts to turn and the ride lasts 180 seconds. Your height h (in feet) above the ground at any time t (in seconds) can be modeled by the equation

$$h(t) = 85 \sin\left[\frac{\pi}{20}(t-6)\right] + 90.$$

a. What is the period?

b. What does the period represent?

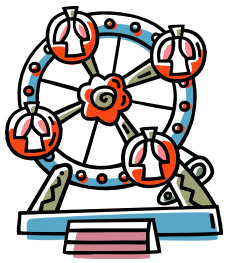
c. What is the frequency?

d. What does the frequency represent?

e. What is your maximum height?

f. What is your minimum height?

g. How many circles will the Ferris Wheel make during the ride?



h. **Calculator active.** How high are you when the ride begins? (Remember, you are not at ground level because the people in line behind you had to get on the ride.)

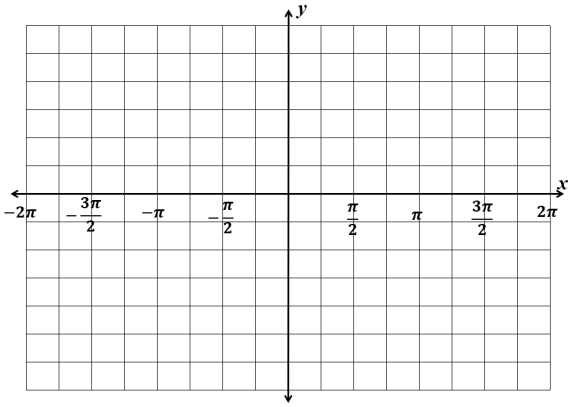
i. **calculator active.** How far off the ground are you when the ride stops?

3.6B Sinusoidal Functions Transformations

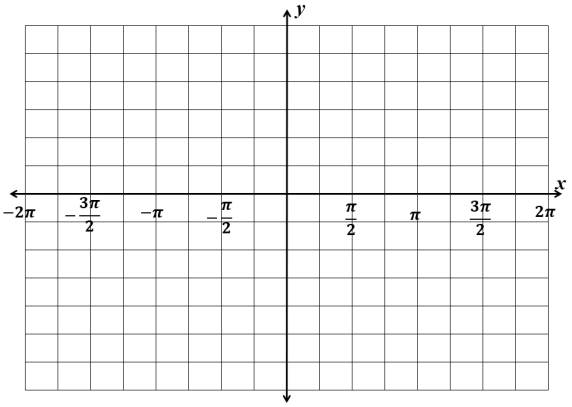
AP Precalculus Name: _____ **CA #1**

Graph the trig function.

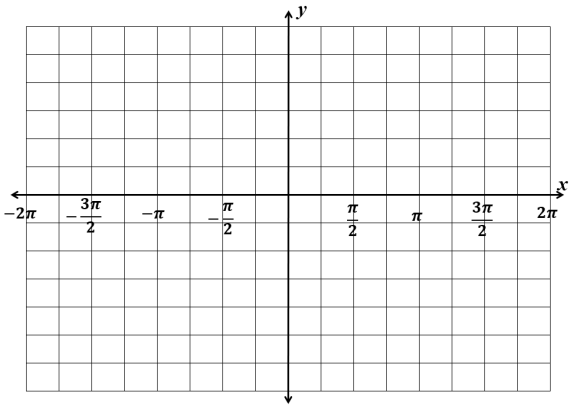
1. $3\cos\left(x-\frac{3\pi}{4}\right)-1$



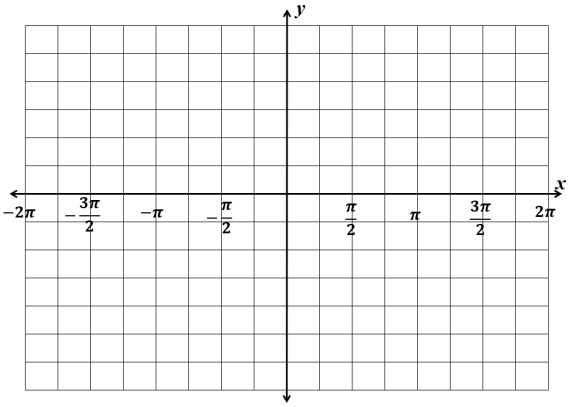
2. $2\sin\left(3\left(x+\frac{\pi}{4}\right)\right)+2$



3. $y=-3\cos\left(2x-\frac{\pi}{4}\right)+1$



4. $y=-\sin(2x+\pi)$



Create a sine function that has the following attributes.

5.

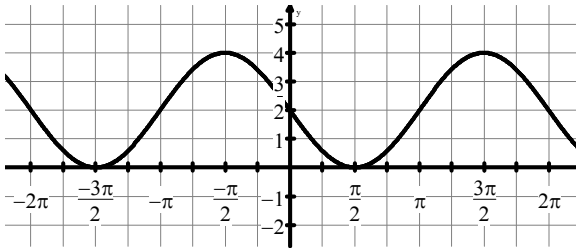
Amplitude: 8
Period: $\frac{\pi}{3}$
Phase Shift: left $\frac{\pi}{9}$
Vertical Shift: up 4

6.

Amplitude: 2
Period: $\frac{5\pi}{6}$
Phase Shift: right $\frac{10\pi}{7}$
Vertical Shift: down 3

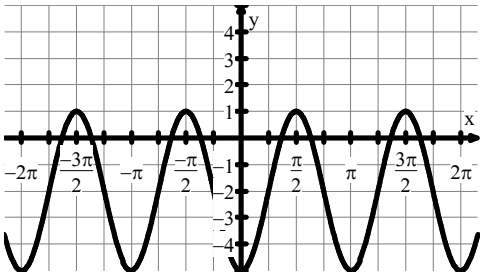
Write the equation of the following *sine* curves.

7.



$y =$ _____

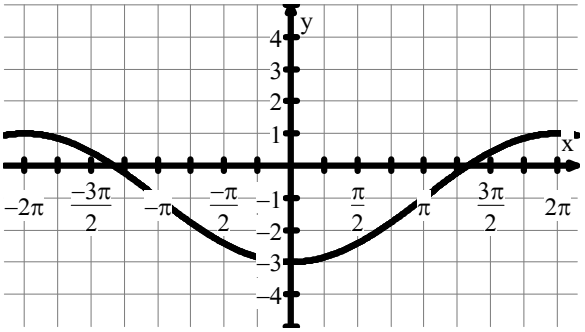
8.



$y =$ _____

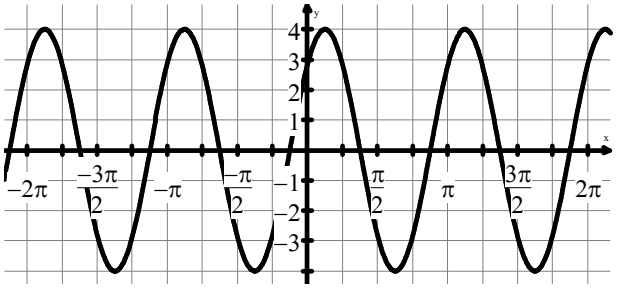
Write the equation of the following *cosine* curves.

9.



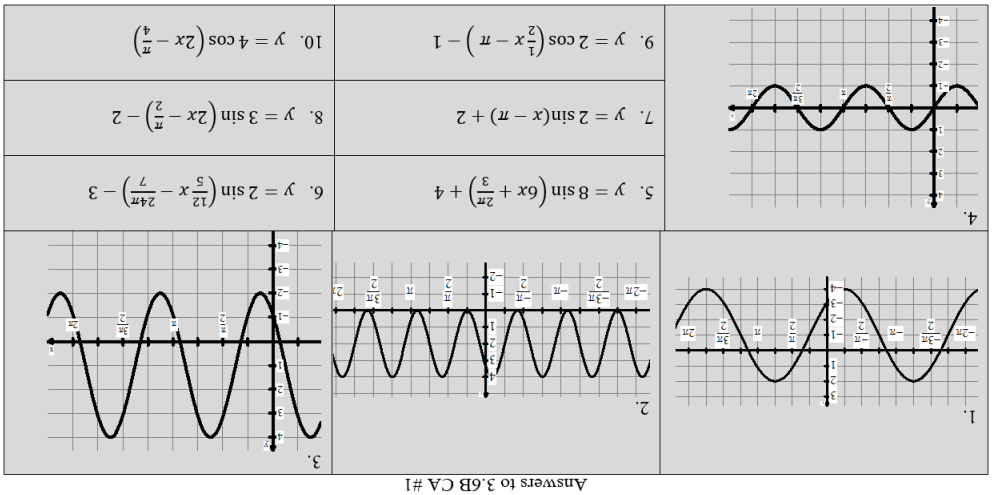
$y =$ _____

10.



$y =$ _____

Answers to 3.6B CA #1



AP Precalc

Write your questions and thoughts here!

3.7 Sinusoidal Function Context and Data Modeling

3.7 Notes

The following data set can be modeled by a sinusoidal function. Use the data to answer each problem below.

x	0	10	20	30	40	50	60	70	80	90
$f(x)$	130	184	225	172	126	98	137	169	224	168

x	100	110	120	130	140	150	160	170	180	190
$f(x)$	132	102	128	181	227	171	129	101	141	183

1. Estimate the period and frequency.

2. Estimate the vertical shift (midline).

3. Estimate the amplitude.

4. Using the above information, create a sinusoidal function model.

5. Using a calculator, find a sinusoidal model from the given data set. Your answer should look similar to your estimate of the model in #4.

These are the types of problems in the practice, but there are many more types of problems that you will see in AP Classroom and on the AP Exam. The test prep section of this lesson packet should give you good exposure to those types of problems.

3.7 Sinusoidal Function Context and Data Modeling

AP Precalculus

3.7 Practice

1. The following data set can be modeled by a sinusoidal function. Use the data to answer each problem below.

x	0	1	2	3	4	5	6	7	8	9
$f(x)$	97	24	-62	18	104	21	-65	16	101	23

x	10	11	12	13	14	15	16	17	18	19
$f(x)$	-59	22	98	17	-60	19	99	18	-61	23

a. Estimate the period and frequency.

b. Estimate the vertical shift (midline).

c. Estimate the amplitude.

d. Using the above information, create a sinusoidal function model.

e. Using a calculator, find a sinusoidal model from the given data set. Your answer should look similar to your estimate in part d.

2. The following data set can be modeled by a sinusoidal function. Use the data to answer each problem below.

x	0	π	2π	3π	4π	5π	6π	7π	8π	9π
$f(x)$	349	220	180	230	345	470	510	465	348	235

x	10π	11π	12π	13π	14π	15π	16π	17π	18π	19π
$f(x)$	207	240	361	458	498	462	358	210	196	215

a. Estimate the period and frequency.

b. Estimate the vertical shift (midline).

c. Estimate the amplitude.